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Jiang et al.

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(45) **Date of Patent: Sep. 9, 2025**

(54) **SEMICONDUCTOR MEMORY STRUCTURE AND METHOD FOR FORMING THE SAME**

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(73) Assignee: **TAIWAN SEMICONDUCTOR MANUFACTURING COMPANY LTD.**, Hsinchu (TW)

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(65) **Prior Publication Data**

US 2023/0060819 A1 Mar. 2, 2023

(51) **Int. Cl.**

H10B 41/27 (2023.01)
H01L 23/522 (2006.01)

(Continued)

(52) **U.S. Cl.**

CPC ***H10B 43/27*** (2023.02); ***H01L 23/5226*** (2013.01); ***H10B 41/10*** (2023.02); ***H10B 41/27*** (2023.02); ***H10B 43/10*** (2023.02)

(58) **Field of Classification Search**

CPC H10B 43/20–27; H10B 43/10; H10B 41/00–70; H10B 51/00–50; H10B 12/30; H10B 12/488; H10B 12/00–50; H10B 10/00–18; H10B 20/40–50; H10B 53/00–50; H10B 63/84–845; H10B 69/00; H10B 41/10; H10B 41/27; H10B 51/10; H10B 51/20; H01L 23/5226; H01L 23/538; H01L 23/5386; H01L 2924/1437; H01L 2924/1436–14369; H01L 2924/1451; H01L 27/092–0928; H01L 28/40–92; G11C 8/14; G11C 11/408; G11C 11/41–419; G11C 11/401–4099; G11C 11/5621–5642;

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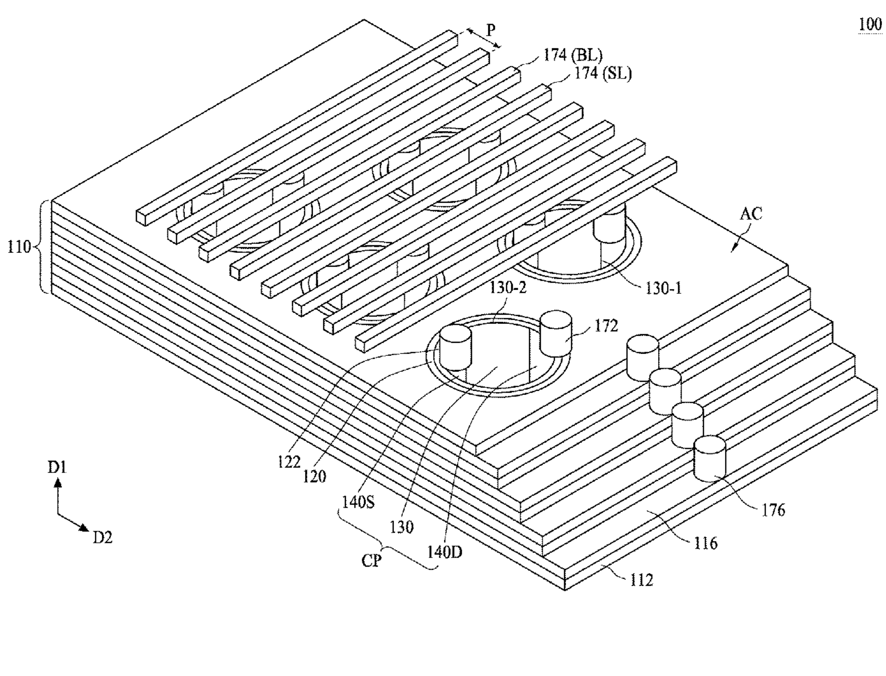
Primary Examiner — David Vu

(74) *Attorney, Agent, or Firm* — WPAT LAW; Anthony King

(57) **ABSTRACT**

A semiconductor memory structure includes a plurality of gate layers and a plurality of insulating layers alternately stacked over a substrate, and at least an active column disposed over the substrate. The gate layers and the insulating layers are alternately stacked along a first direction. The active column extends along the first direction and penetrates the gate layer and the insulating layer. The active column includes a central portion, a charge-trapping layer surrounding the central portion, and a channel layer between the central portion and the charge-trapping layer. The central portion of the active column includes an isolation structure, a source structure and a drain structure. The source structure and the drain structure are disposed at two sides of the isolation structure.

20 Claims, 27 Drawing Sheets



(51) **Int. Cl.**
H10B 41/10 (2023.01)
H10B 43/10 (2023.01)
H10B 43/27 (2023.01)

(58) **Field of Classification Search**
CPC G11C 14/0009–0045; G11C 2211/4016; G11C 16/00–349; G11C 2216/06–10; G11C 2216/12–30
See application file for complete search history.

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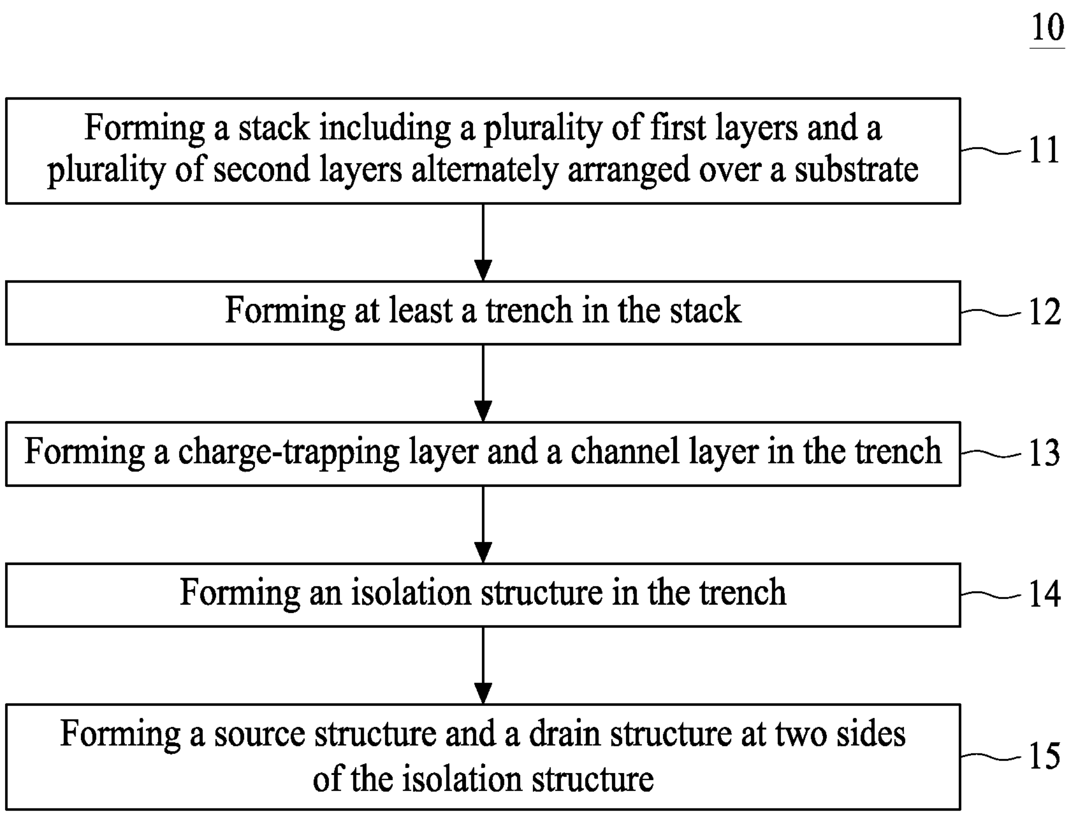
* cited by examiner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Forming a stack including a plurality of first layers and a plurality of second layers alternately arranged over a substrate
11
Forming at least a trench in the stack
12
Forming a charge-trapping layer and a channel layer in the trench
13
Forming an isolation structure in the trench
14
Forming a source structure and a drain structure at two sides of the isolation structure
15

FIG. 1

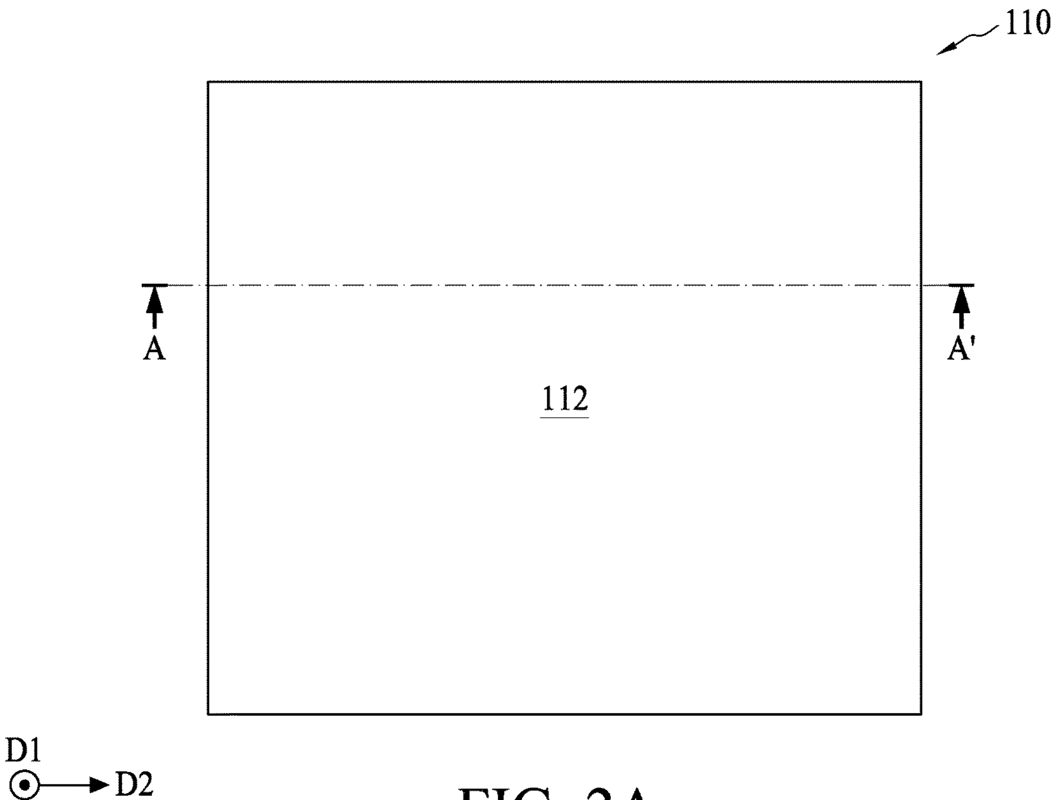


FIG. 2A

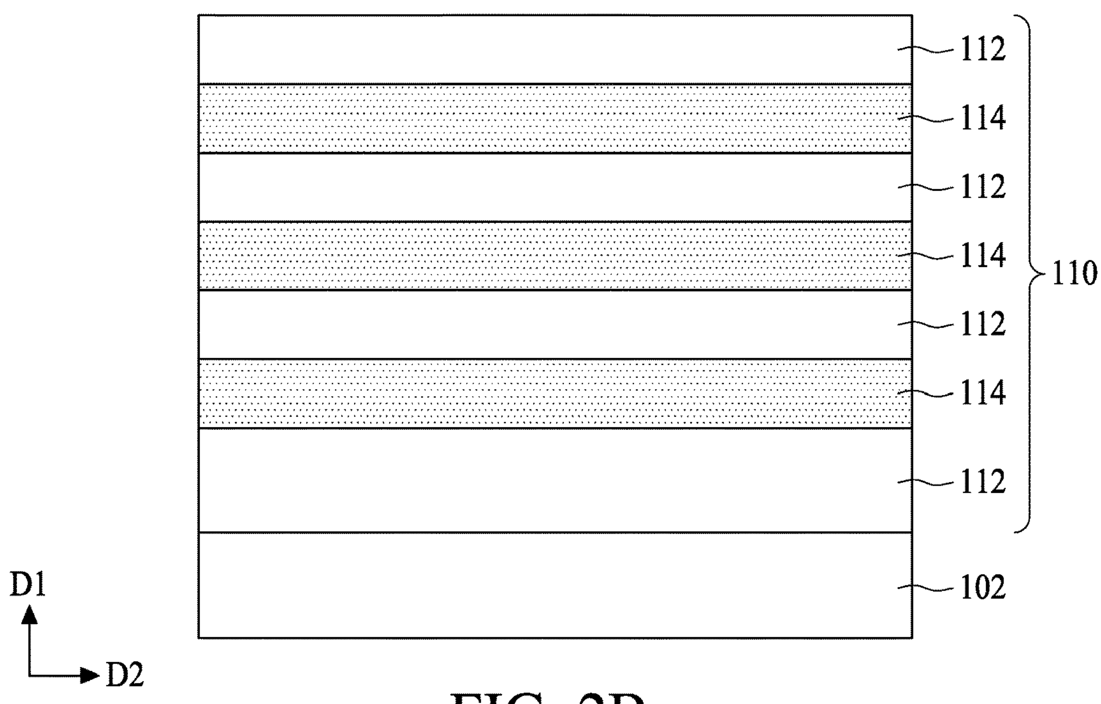


FIG. 2B

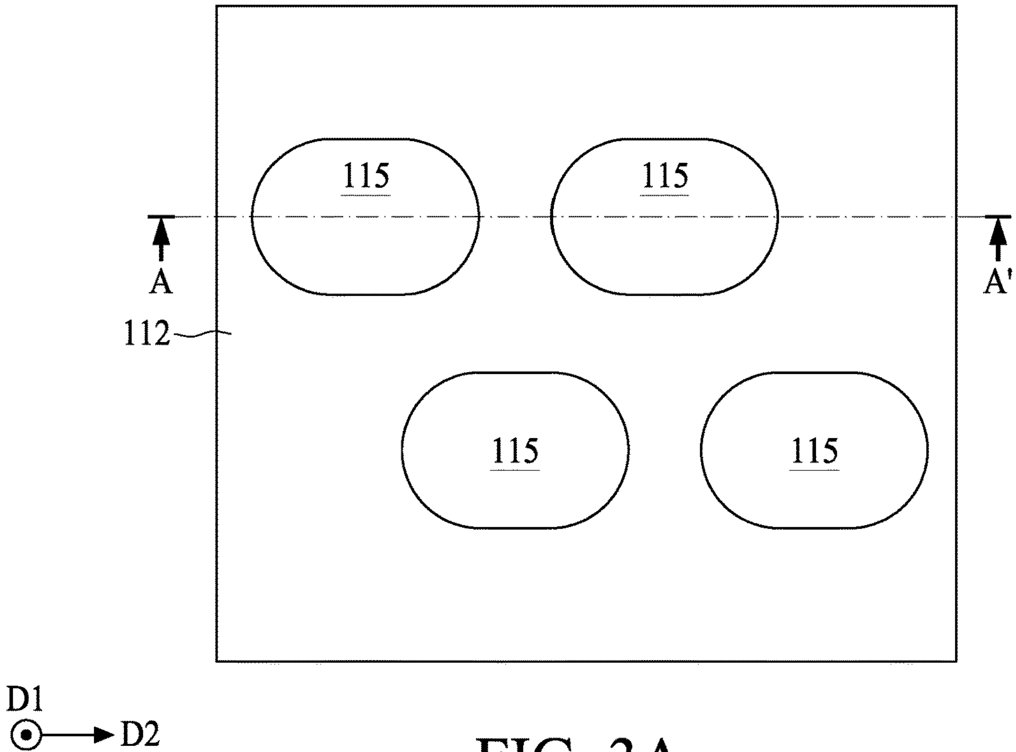
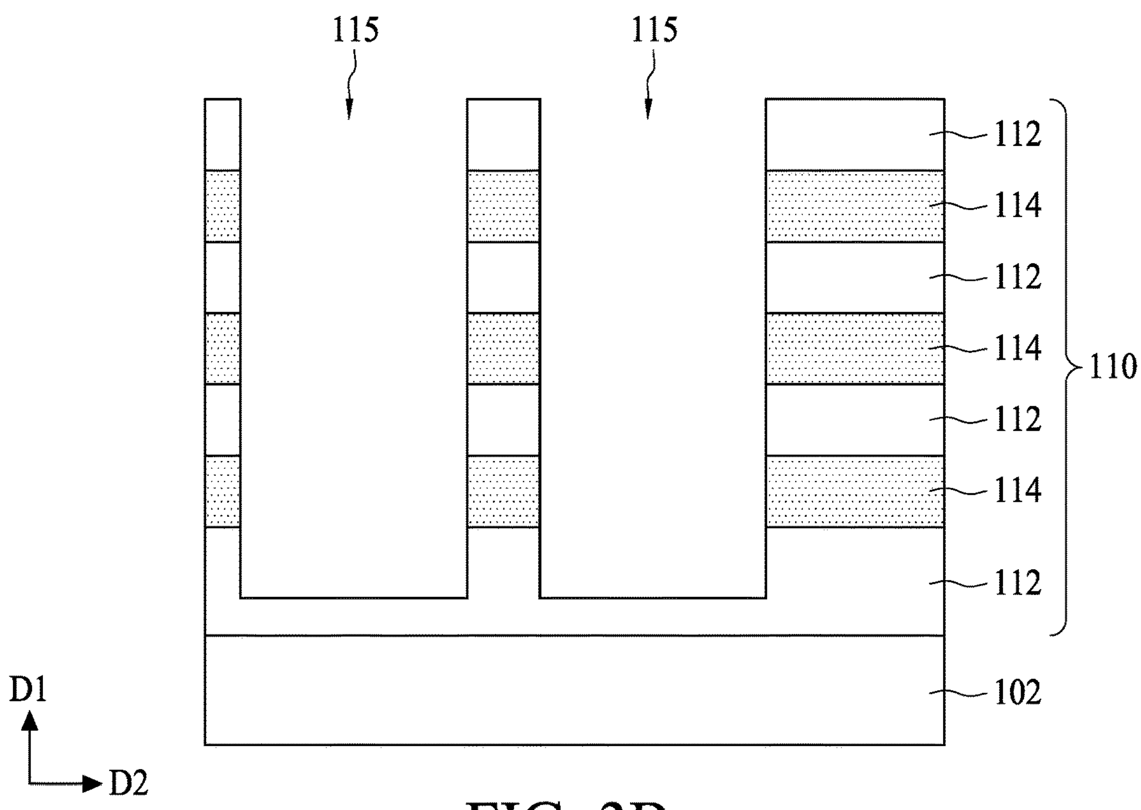

115
115
A
A'
112
115
115
D1
D2

FIG. 3A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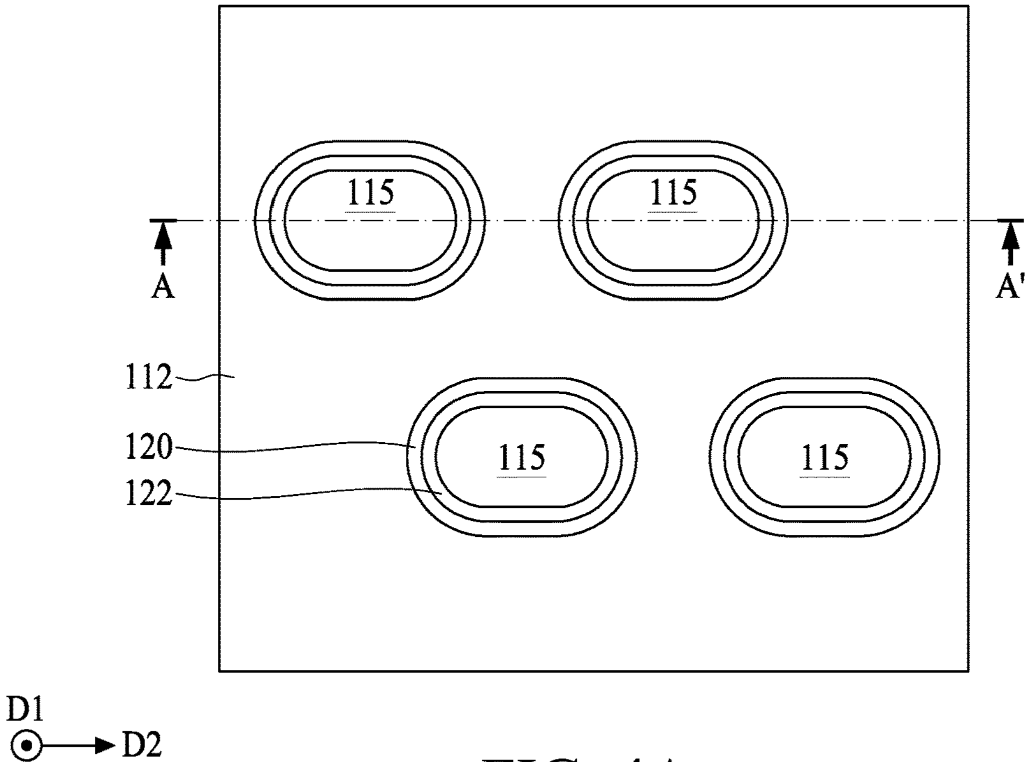
112
102
D1
D2

FIG. 3B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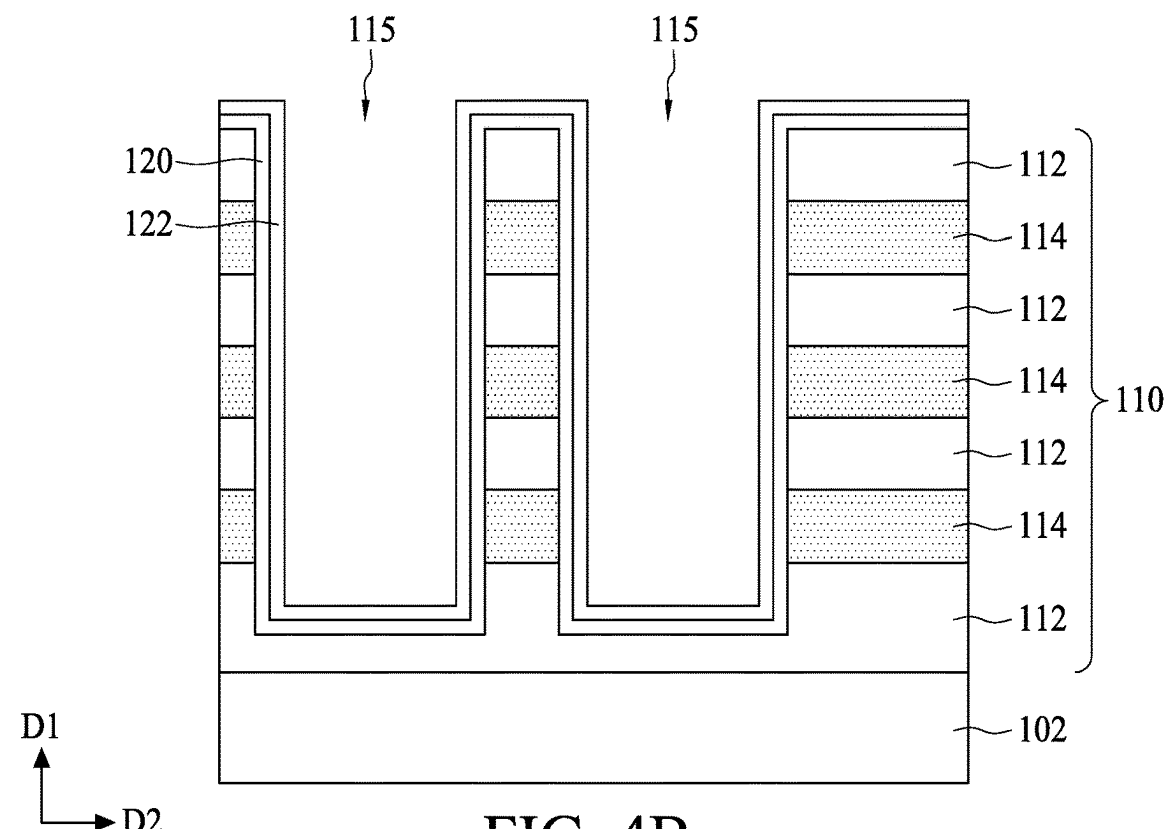
115
A
A'
112
120
122
115
115
D1
D2

FIG. 4A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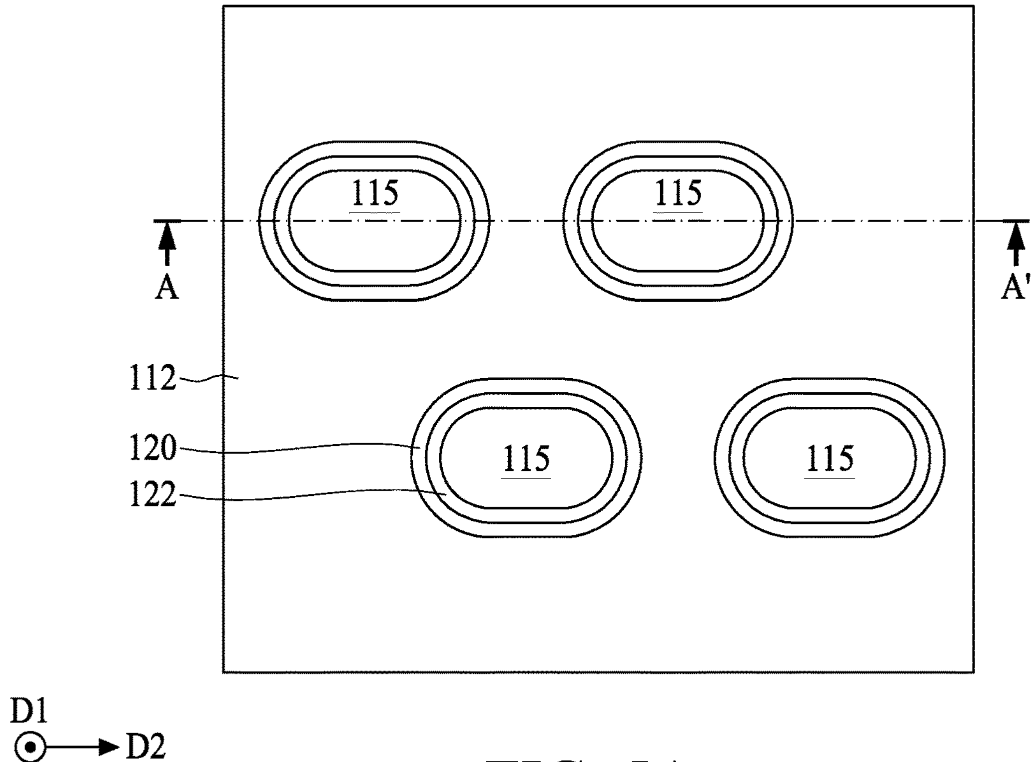
122
112
114
112
114
112
114
112
110
102
D1
D2

FIG. 4B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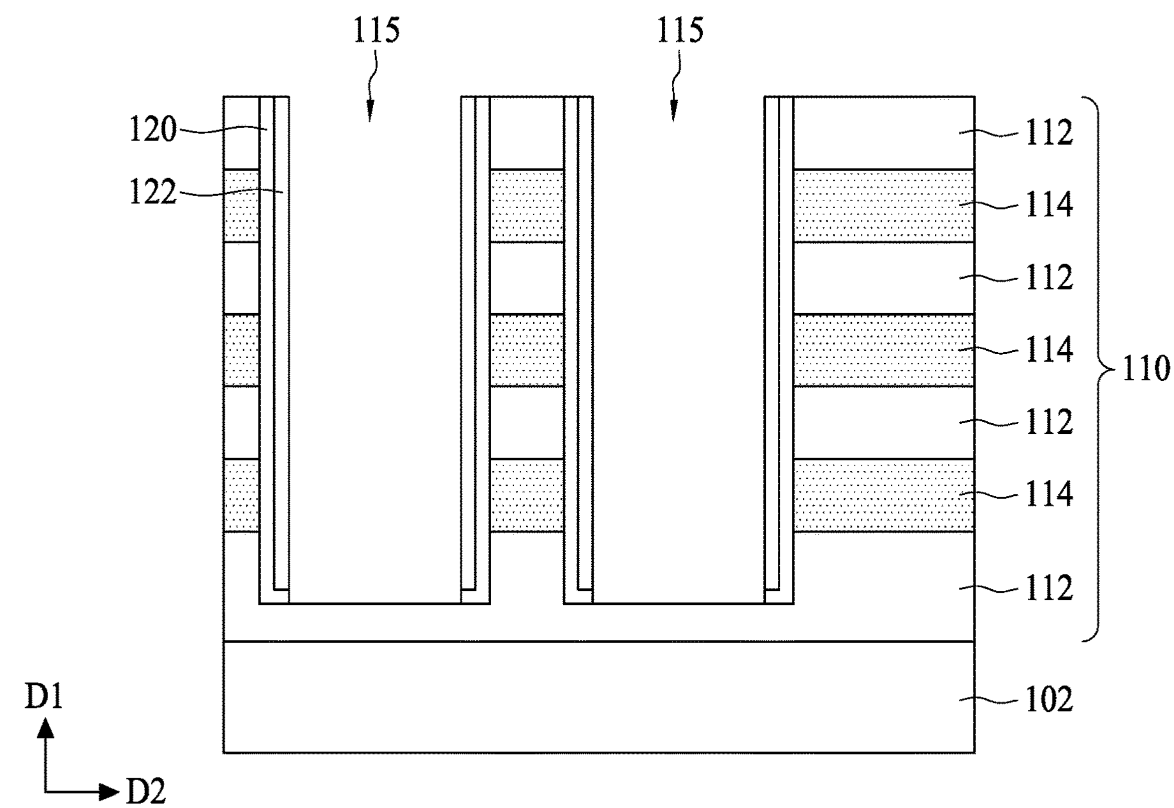
115
A
A'
112
120
122
115
115
D1
D2

FIG. 5A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120
122
112
114
112
114
112
114
112
110
102
D1
D2

FIG. 5B

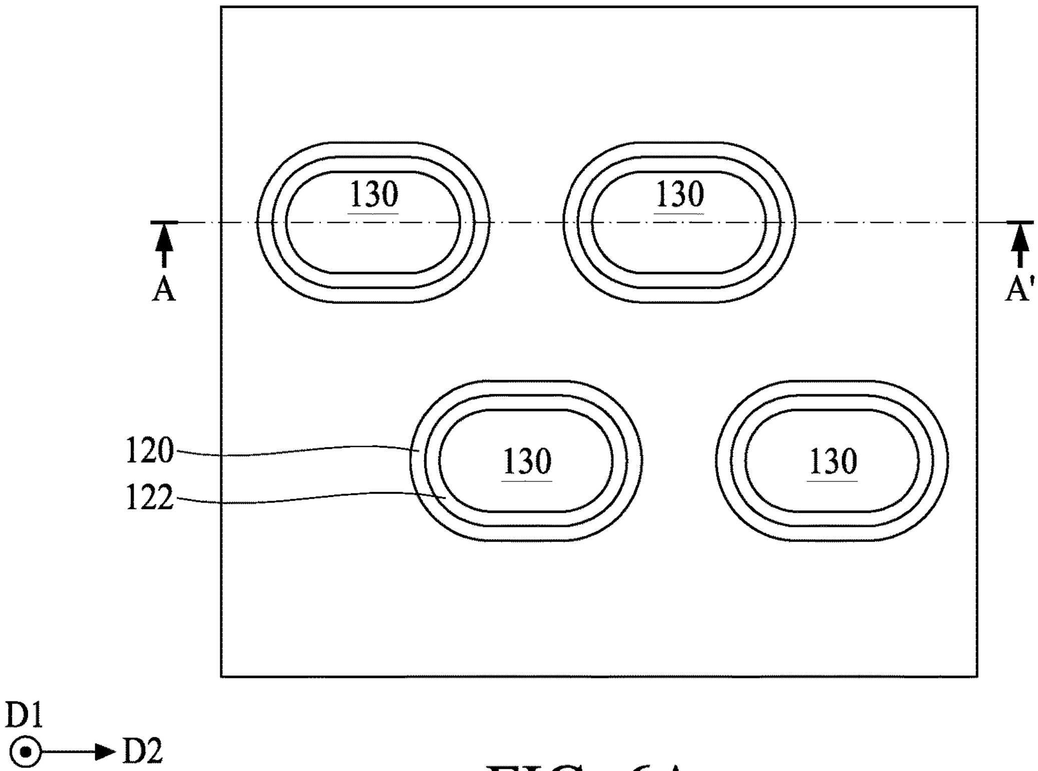
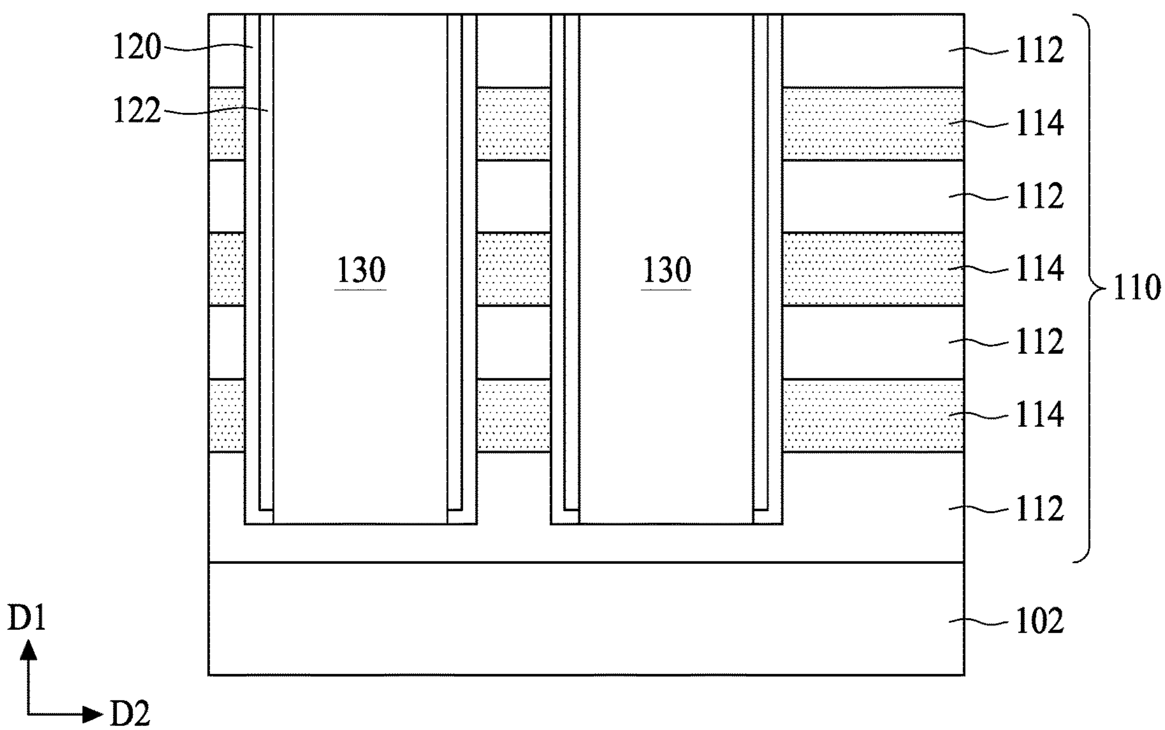

130
130
A
A'
120
122
130
130
D1
D2

FIG. 6A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130
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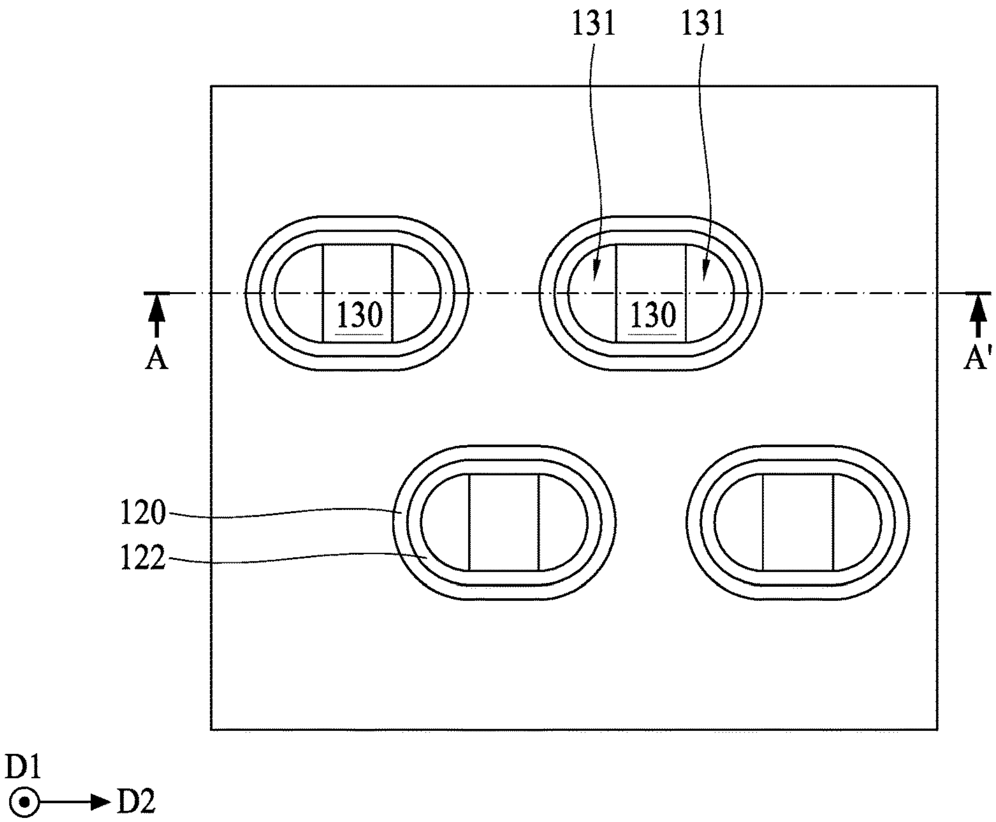
112
114
112
110
102
D1
D2

FIG. 6B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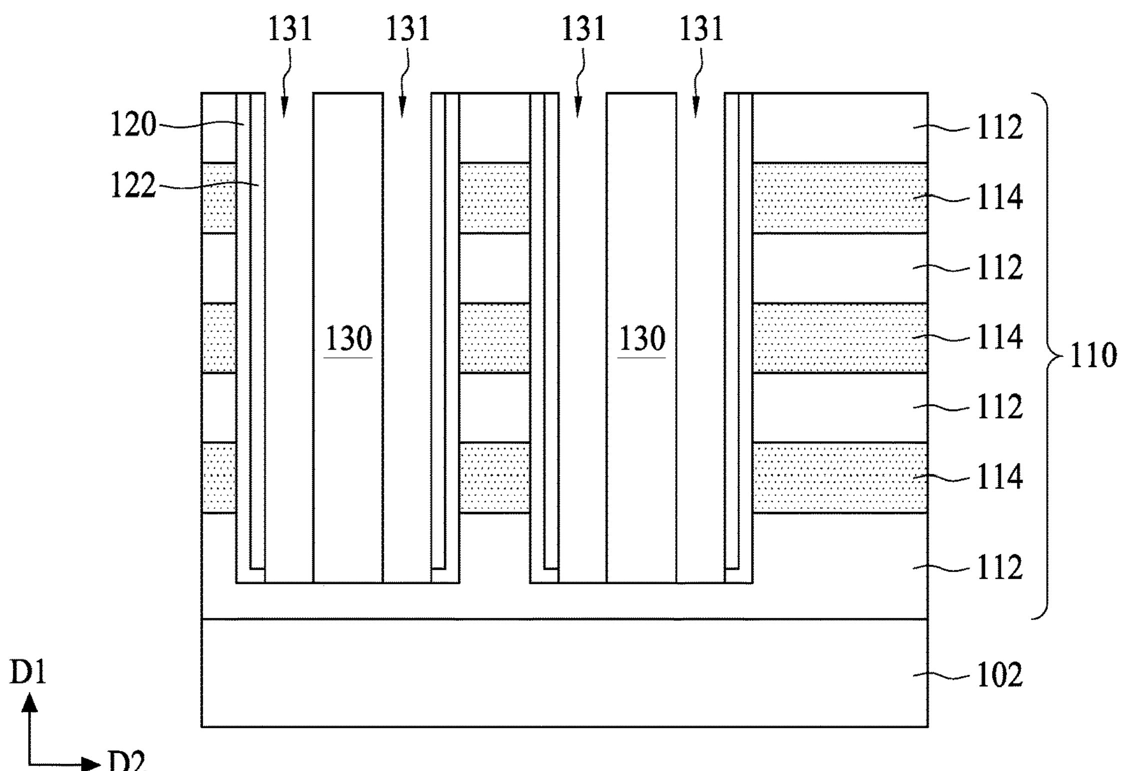
131
130
130
A
A'
120
122
D1
D2

FIG. 7A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130
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114
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110
102
D1
D2

FIG. 7B

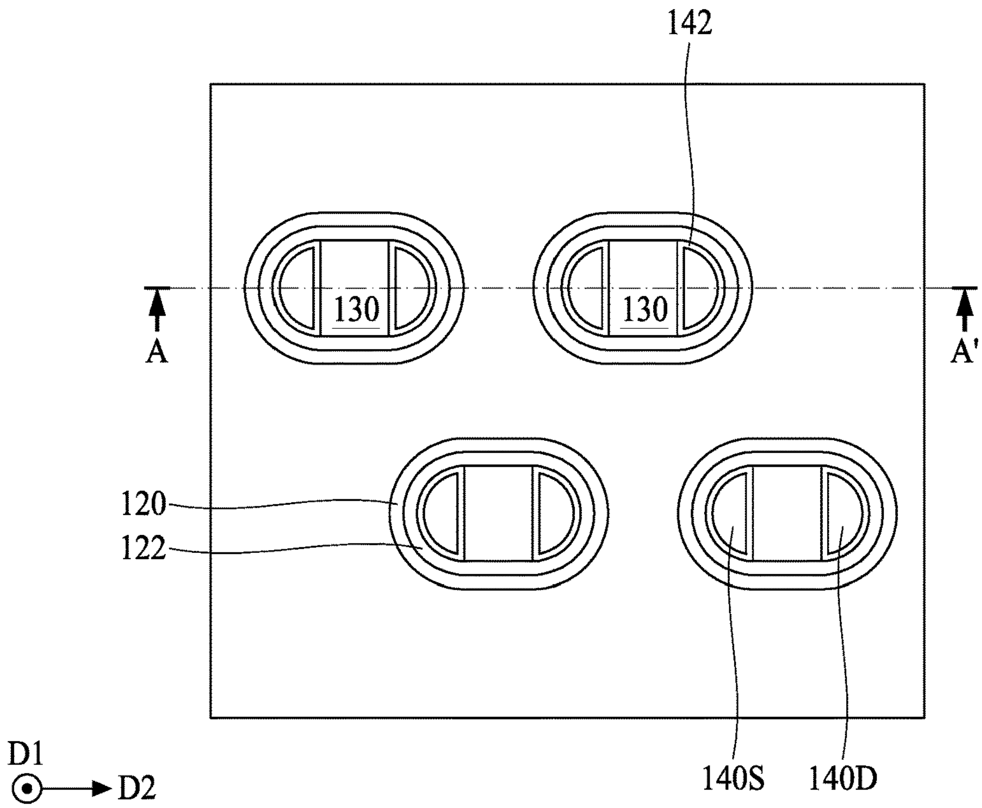
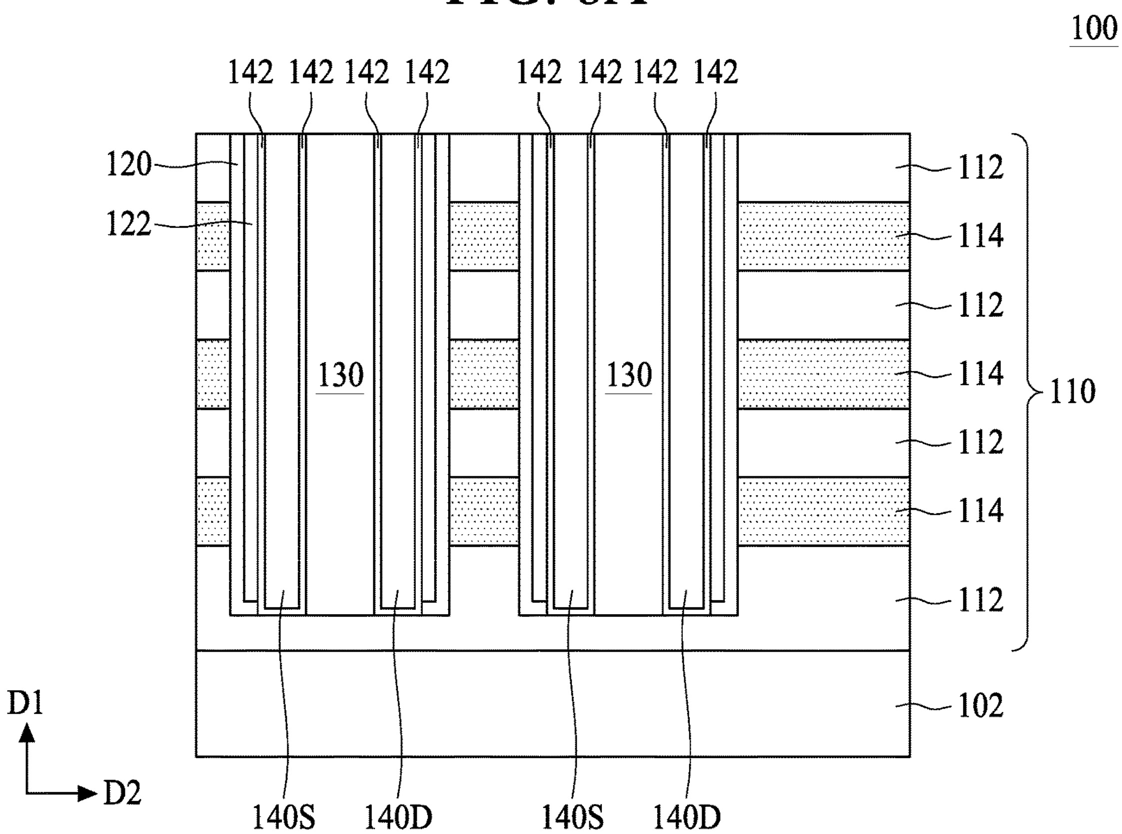

142
130
130
A
A'
120
122
140S
140D
D1
D2

FIG. 8A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140S
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140S
140D
D1
D2

FIG. 8B

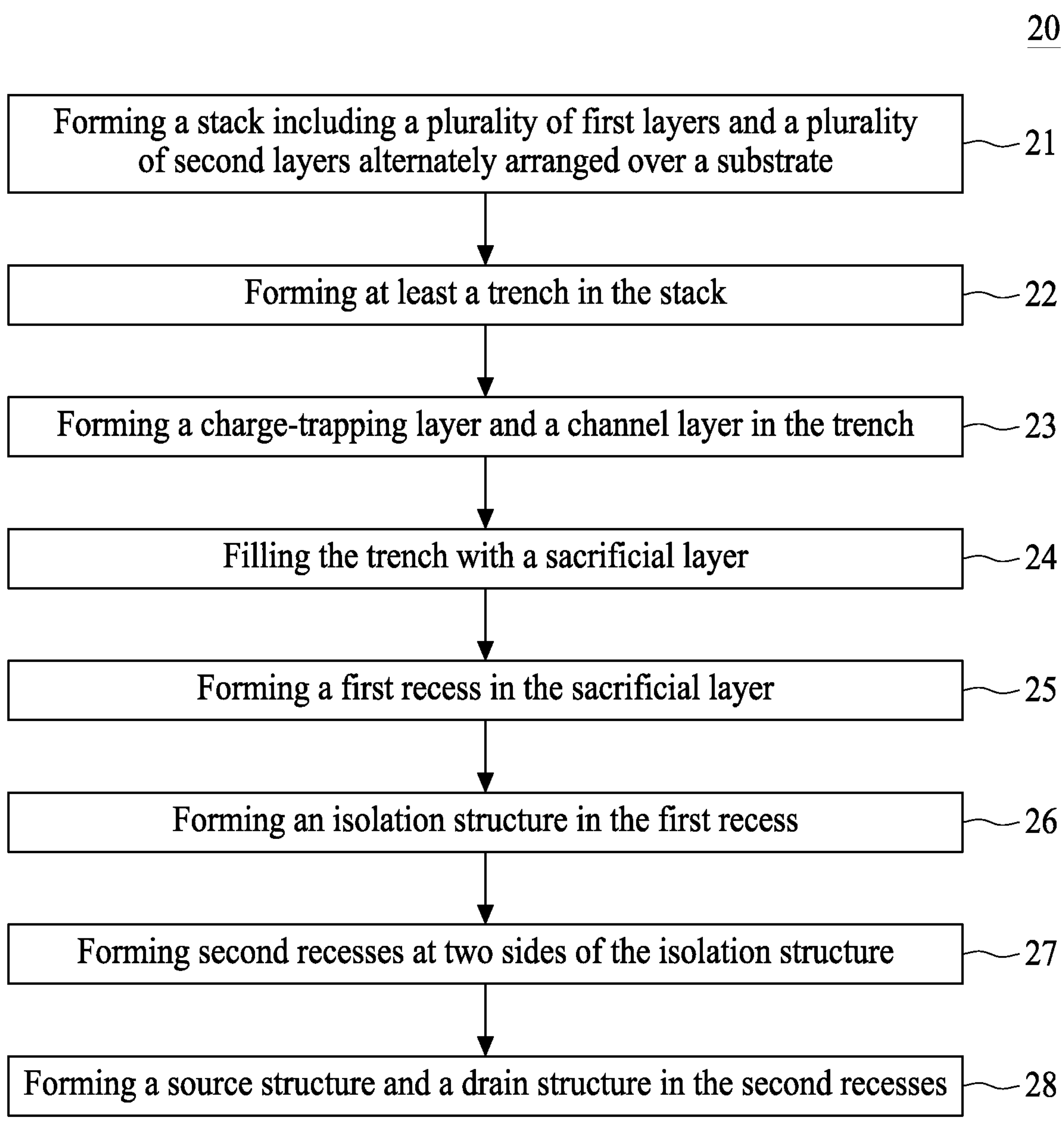

20
Forming a stack including a plurality of first layers and a plurality of second layers alternately arranged over a substrate
21
Forming at least a trench in the stack
22
Forming a charge-trapping layer and a channel layer in the trench
23
Filling the trench with a sacrificial layer
24
Forming a first recess in the sacrificial layer
25
Forming an isolation structure in the first recess
26
Forming second recesses at two sides of the isolation structure
27
Forming a source structure and a drain structure in the second recesses
28

FIG. 9

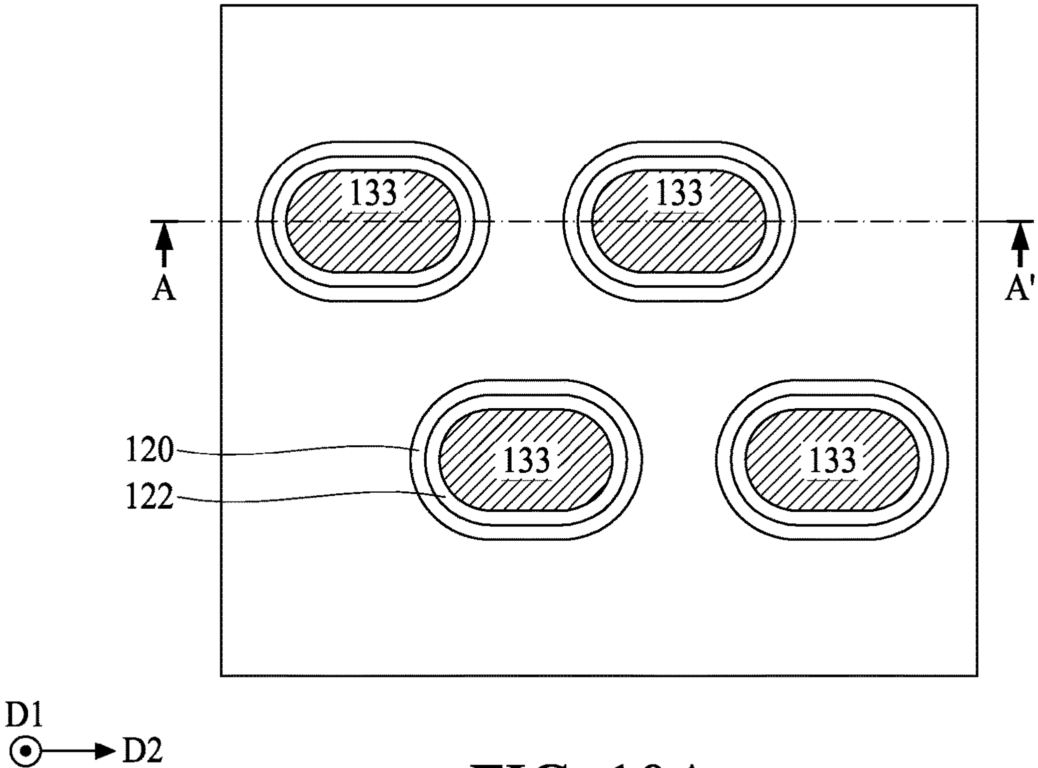


FIG. 10A

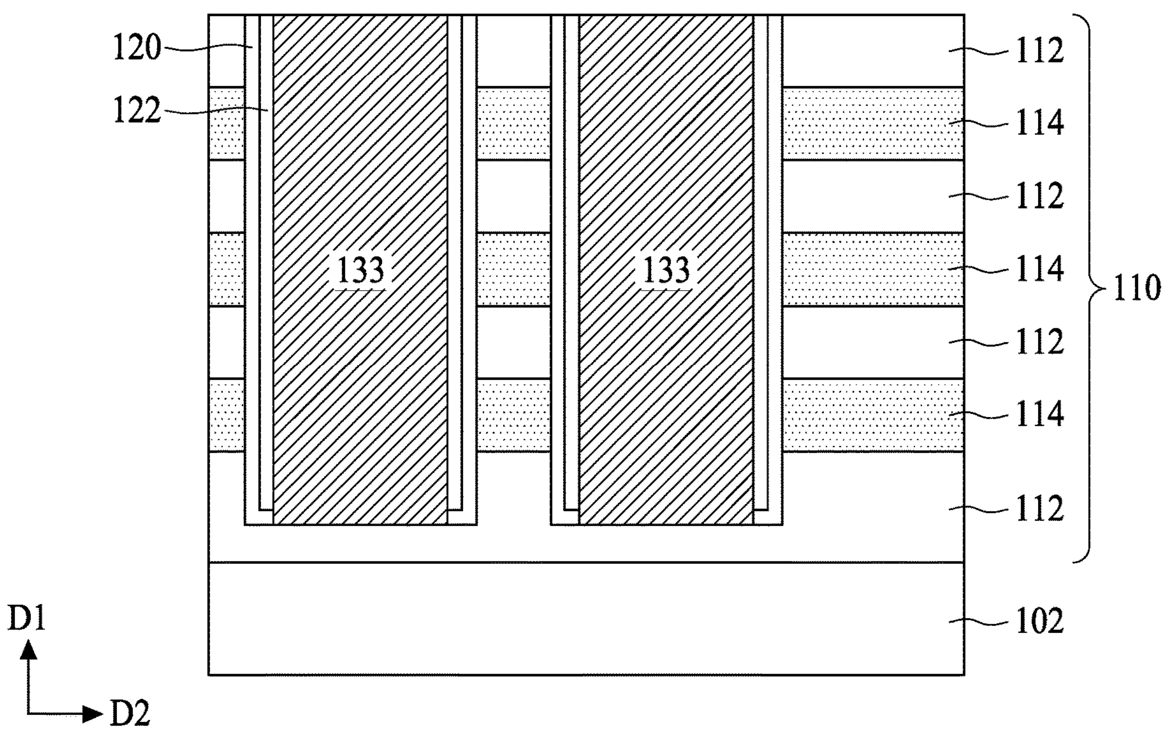


FIG. 10B

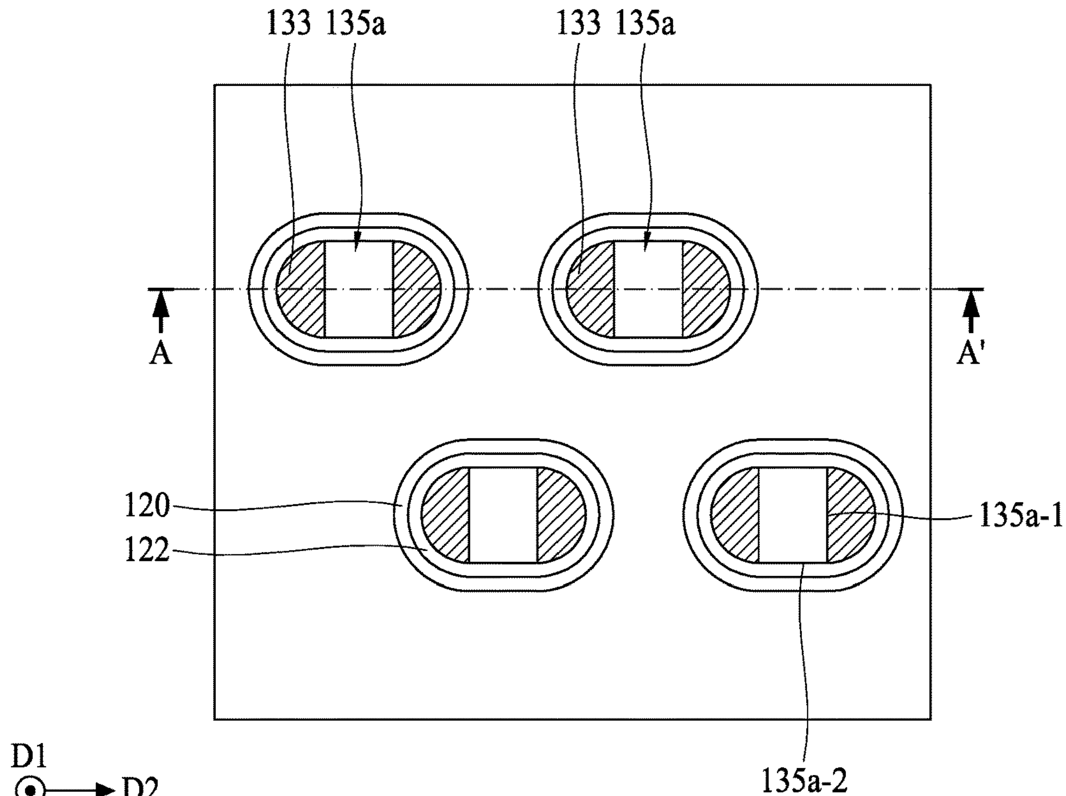


FIG. 11A

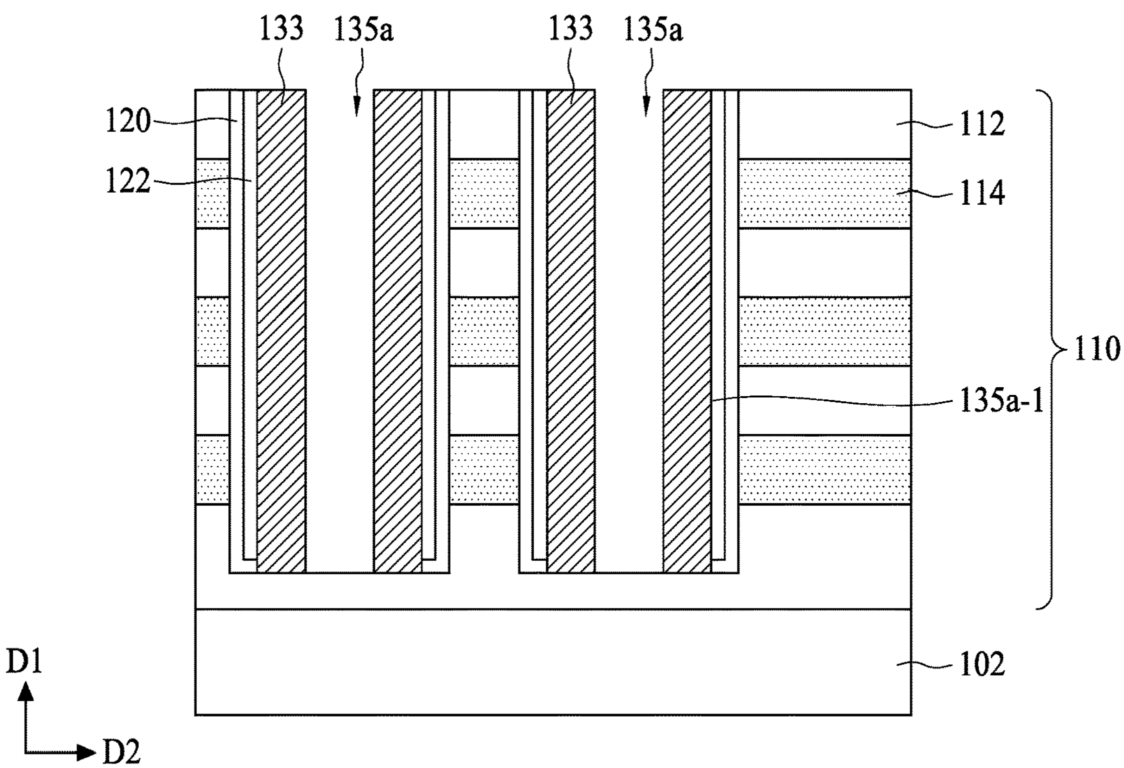


FIG. 11B

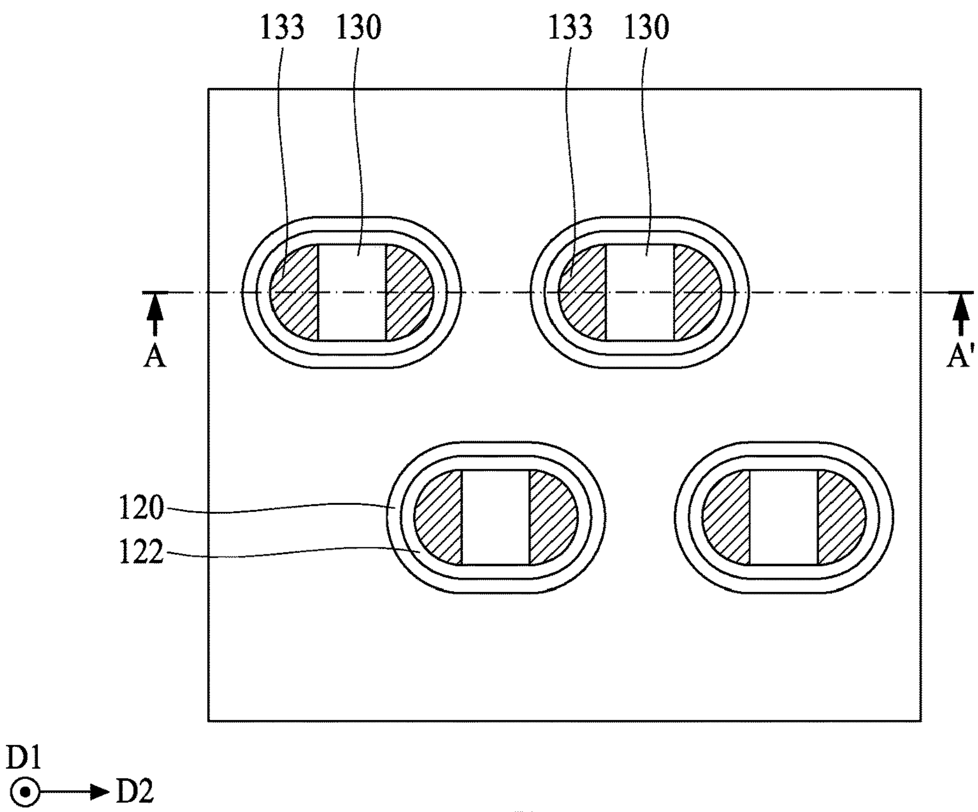


FIG. 12A

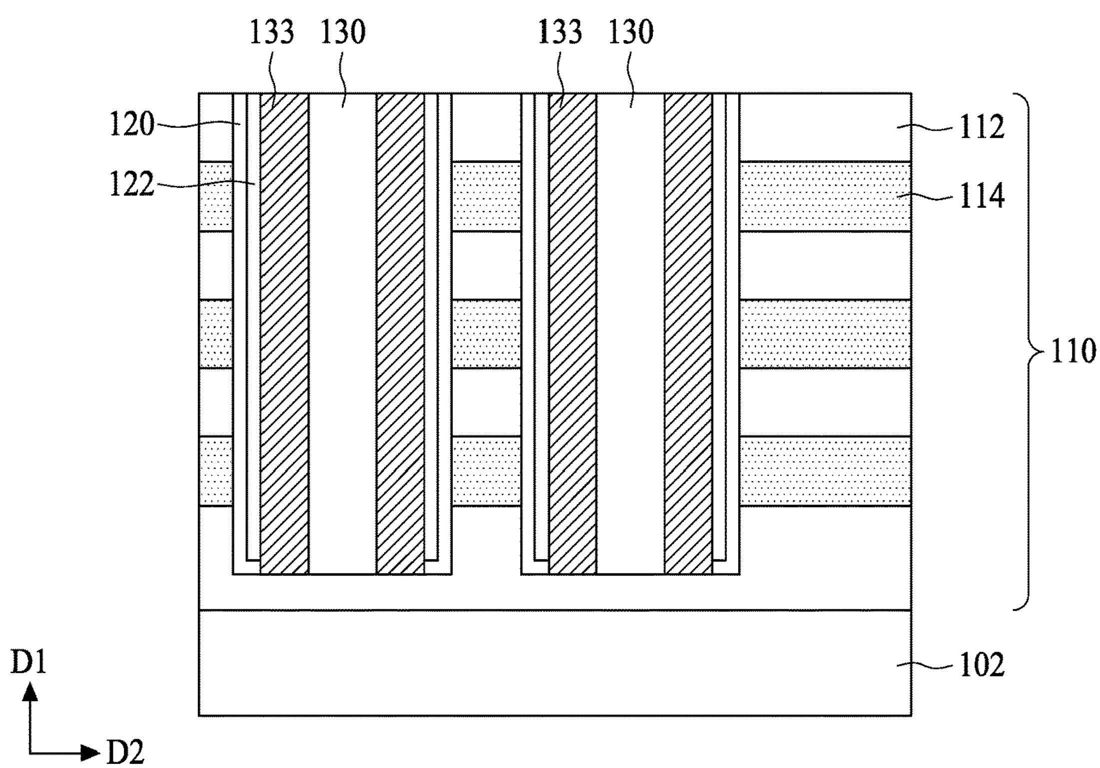


FIG. 12B

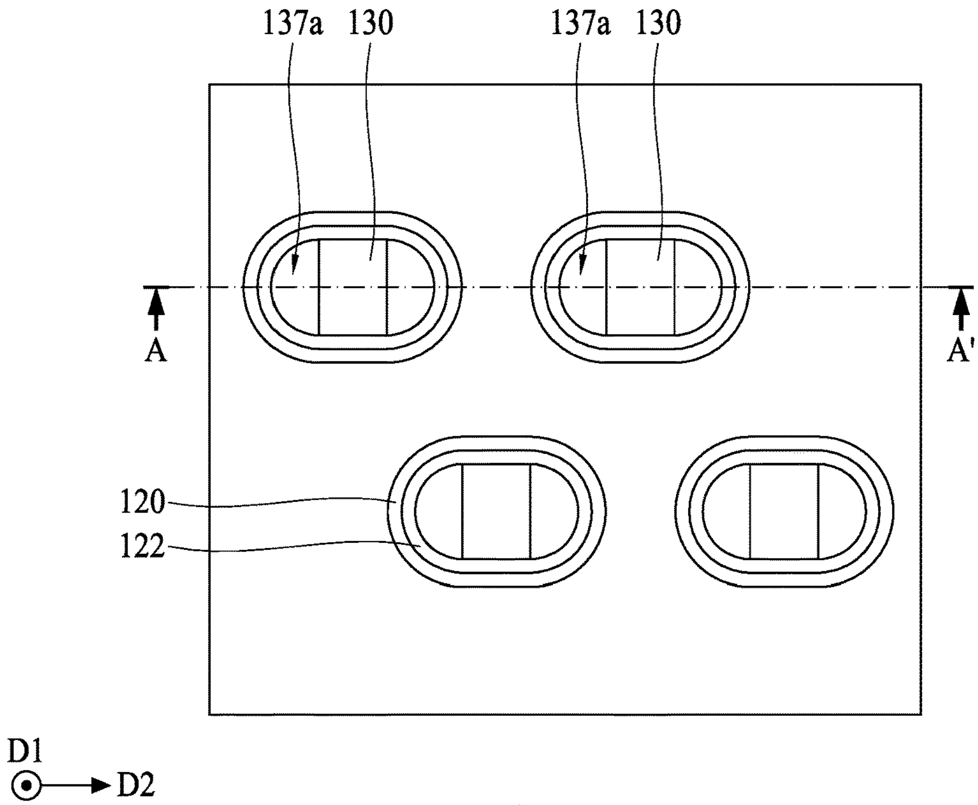


FIG. 13A

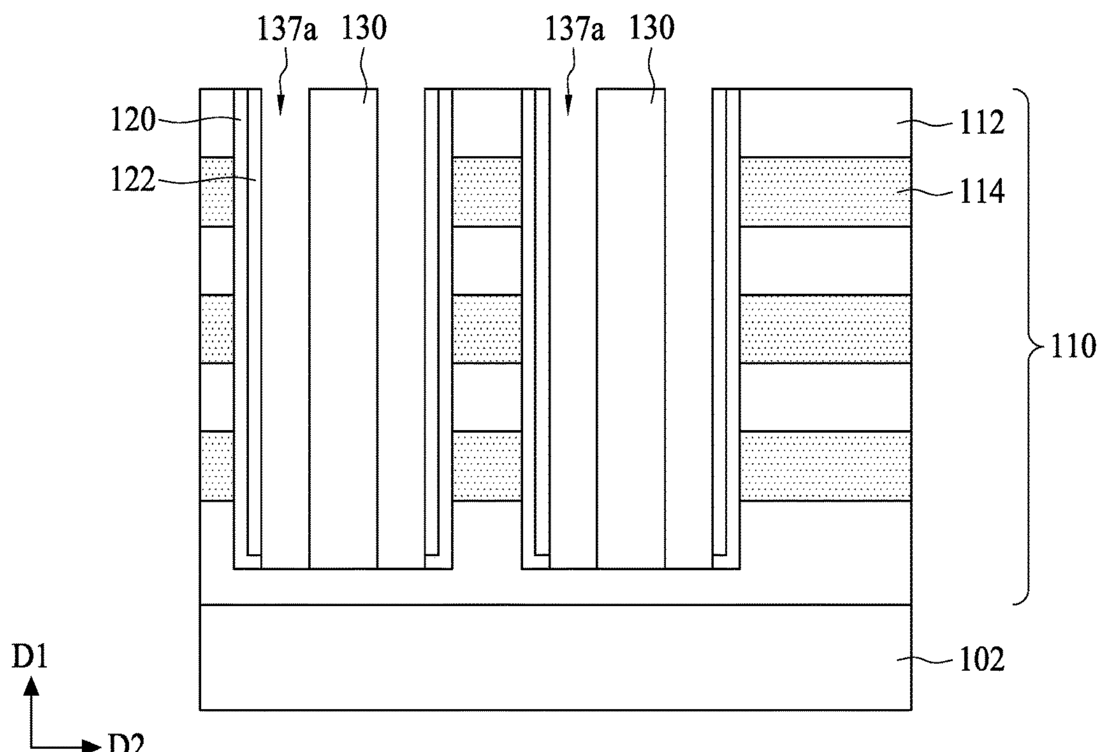


FIG. 13B

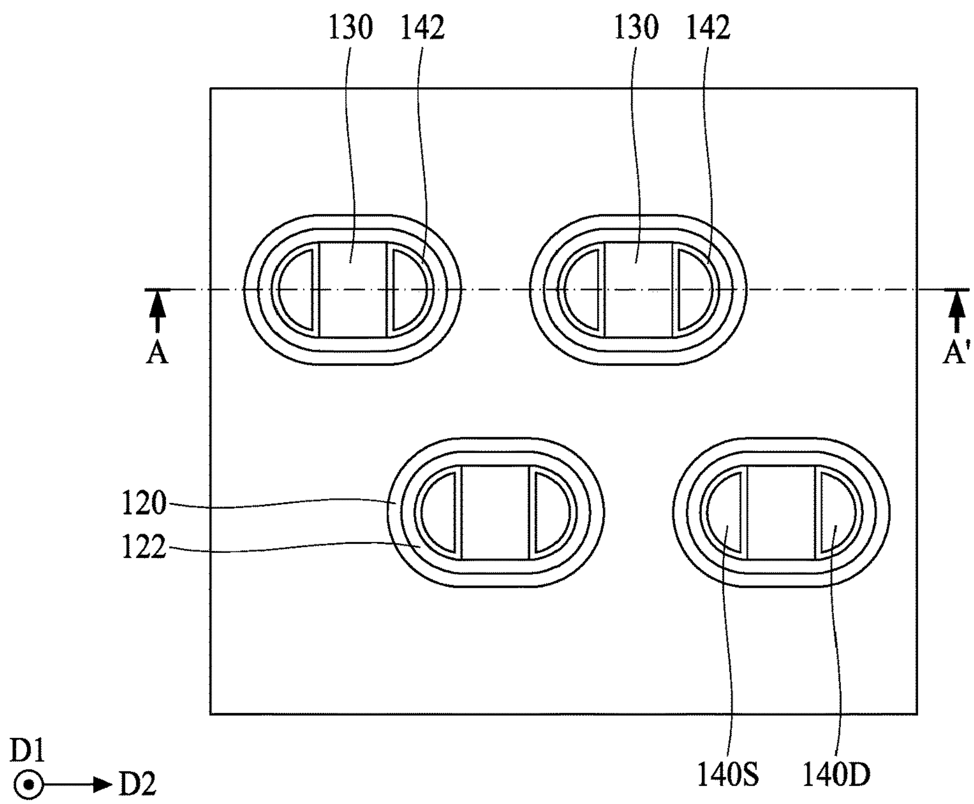


FIG. 14A

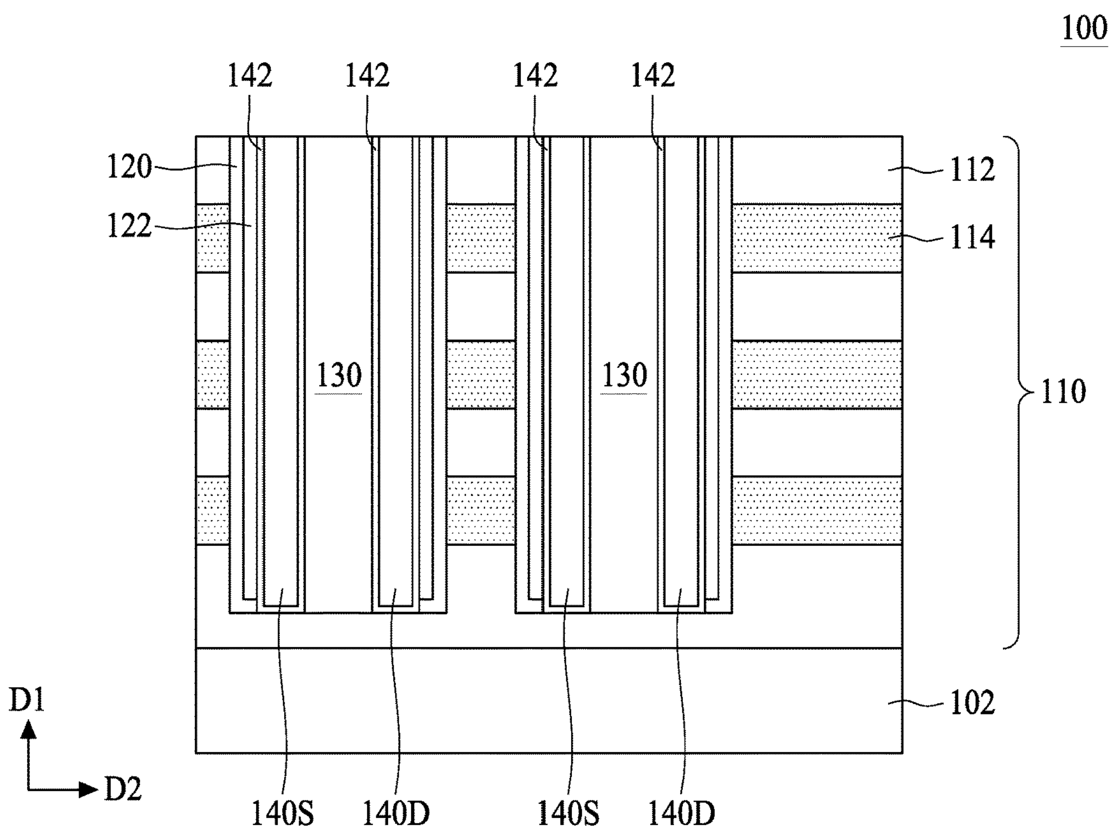


FIG. 14B

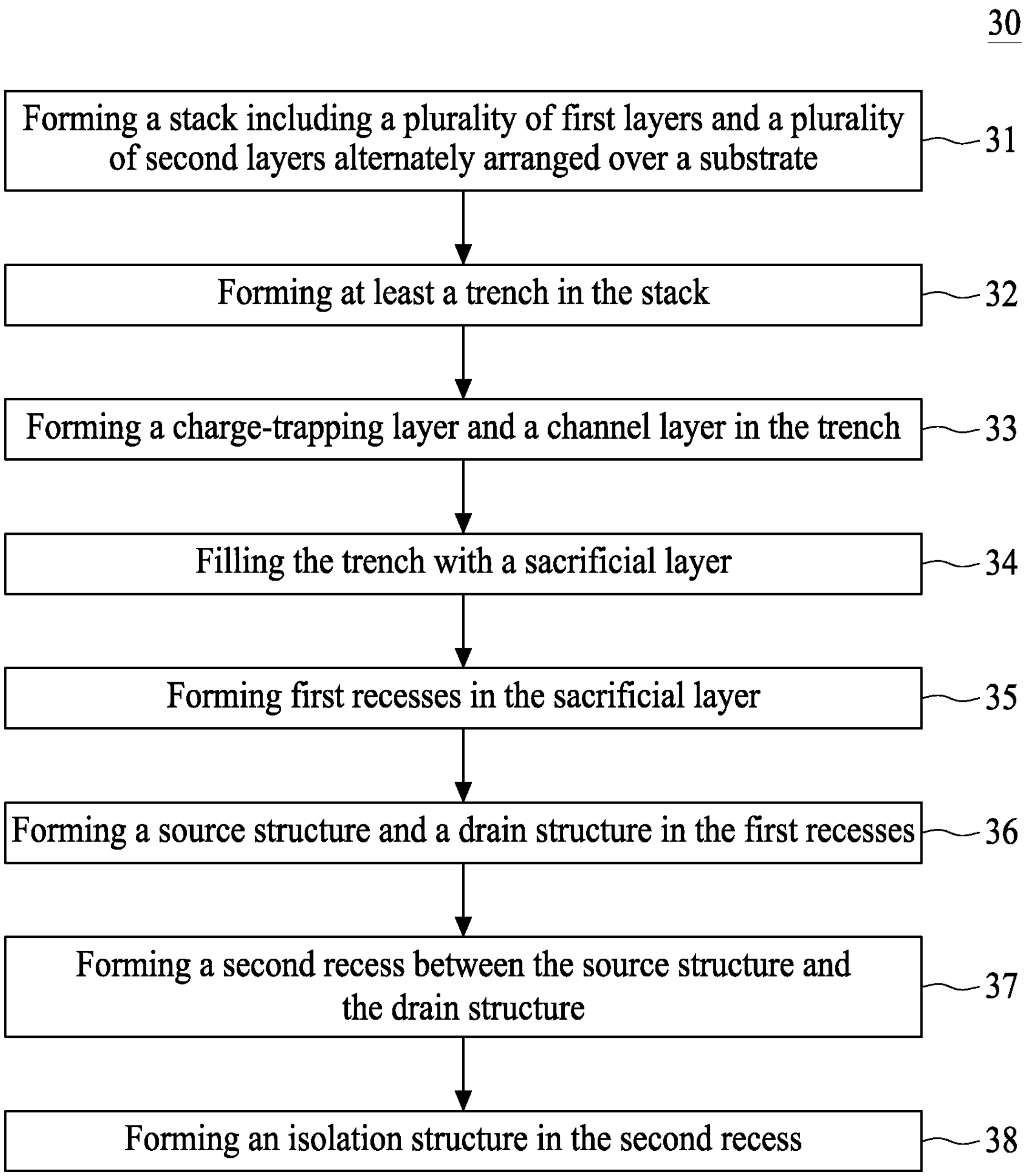

30
Forming a stack including a plurality of first layers and a plurality of second layers alternately arranged over a substrate
31
Forming at least a trench in the stack
32
Forming a charge-trapping layer and a channel layer in the trench
33
Filling the trench with a sacrificial layer
34
Forming first recesses in the sacrificial layer
35
Forming a source structure and a drain structure in the first recesses
36
Forming a second recess between the source structure and the drain structure
37
Forming an isolation structure in the second recess
38

FIG. 15

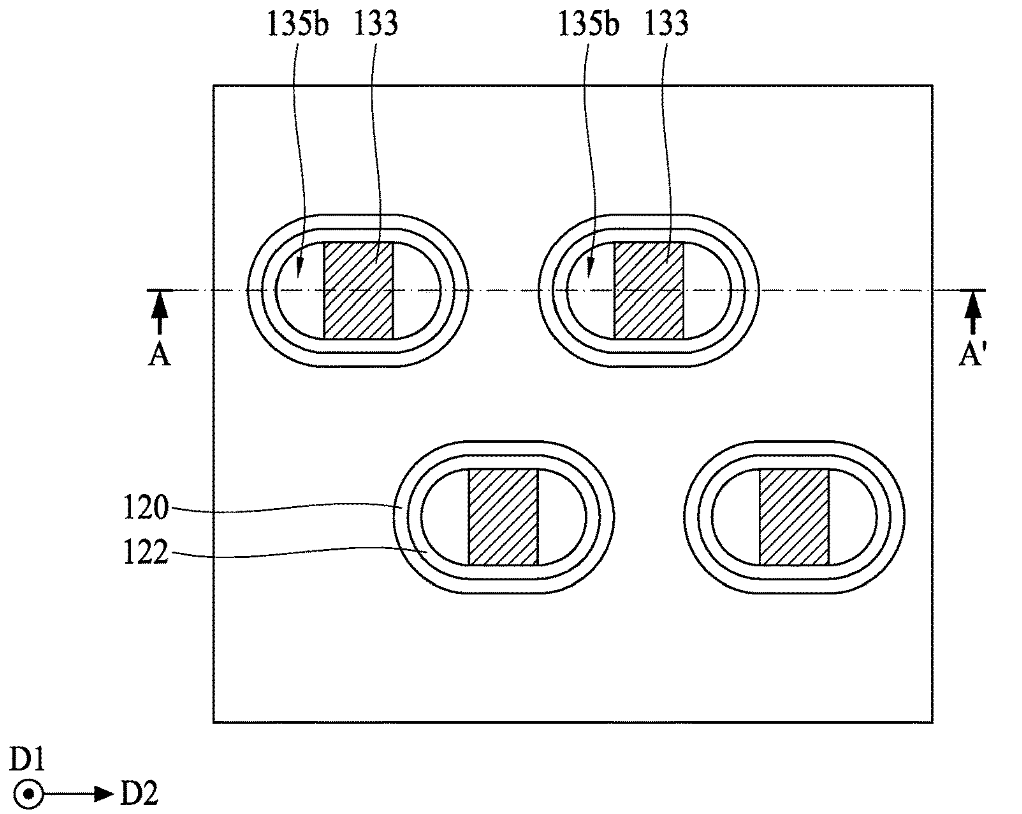


FIG. 16A

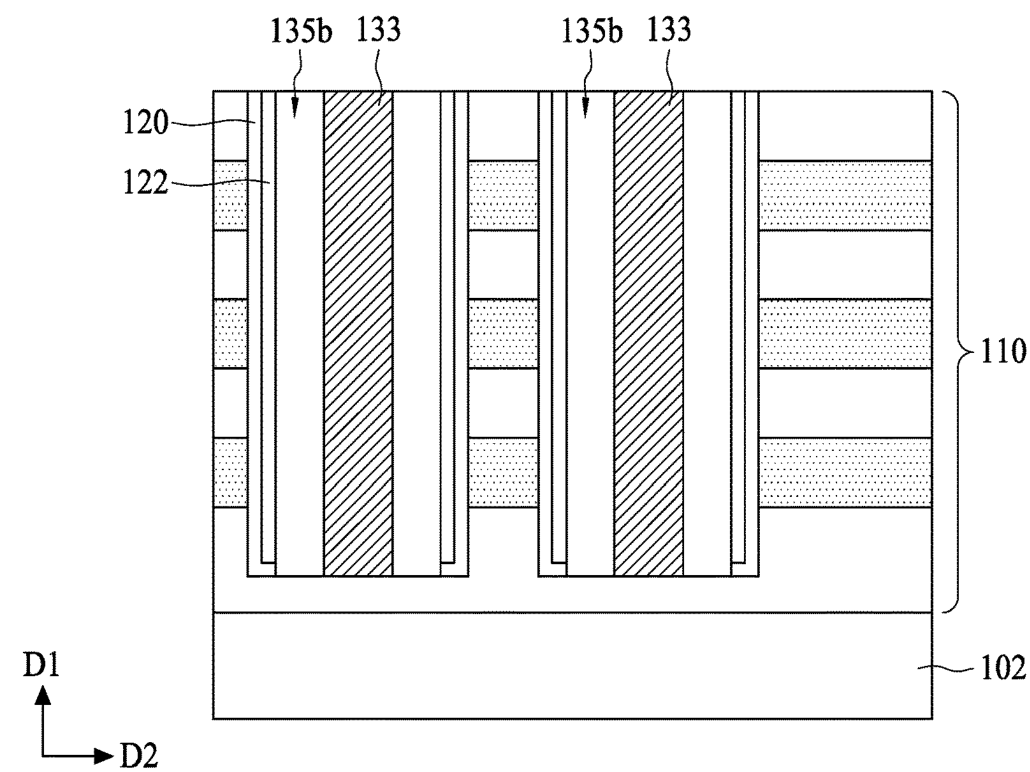


FIG. 16B

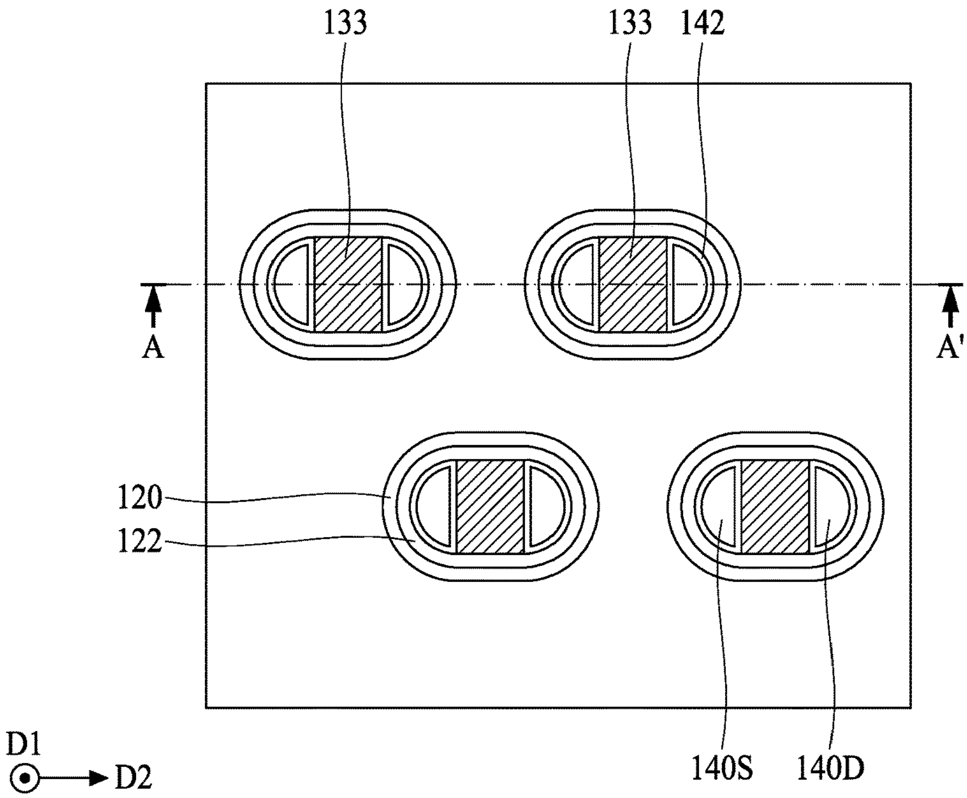


FIG. 17A

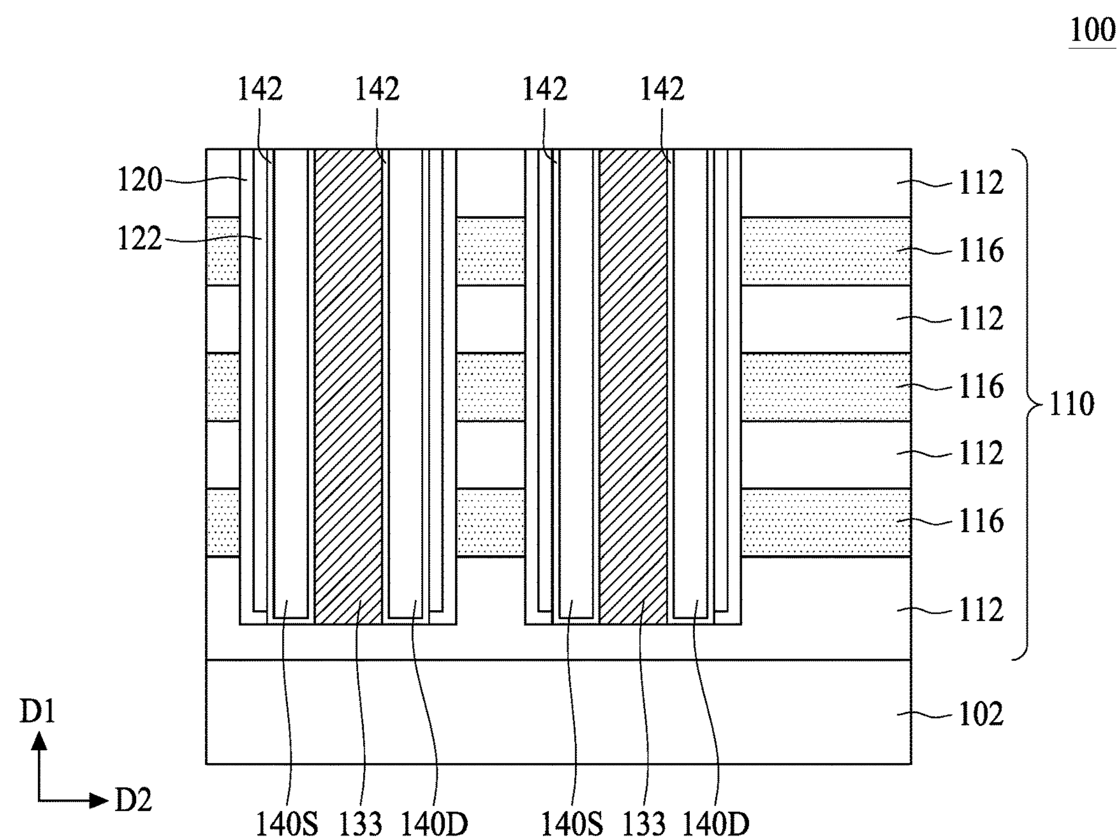


FIG. 17B

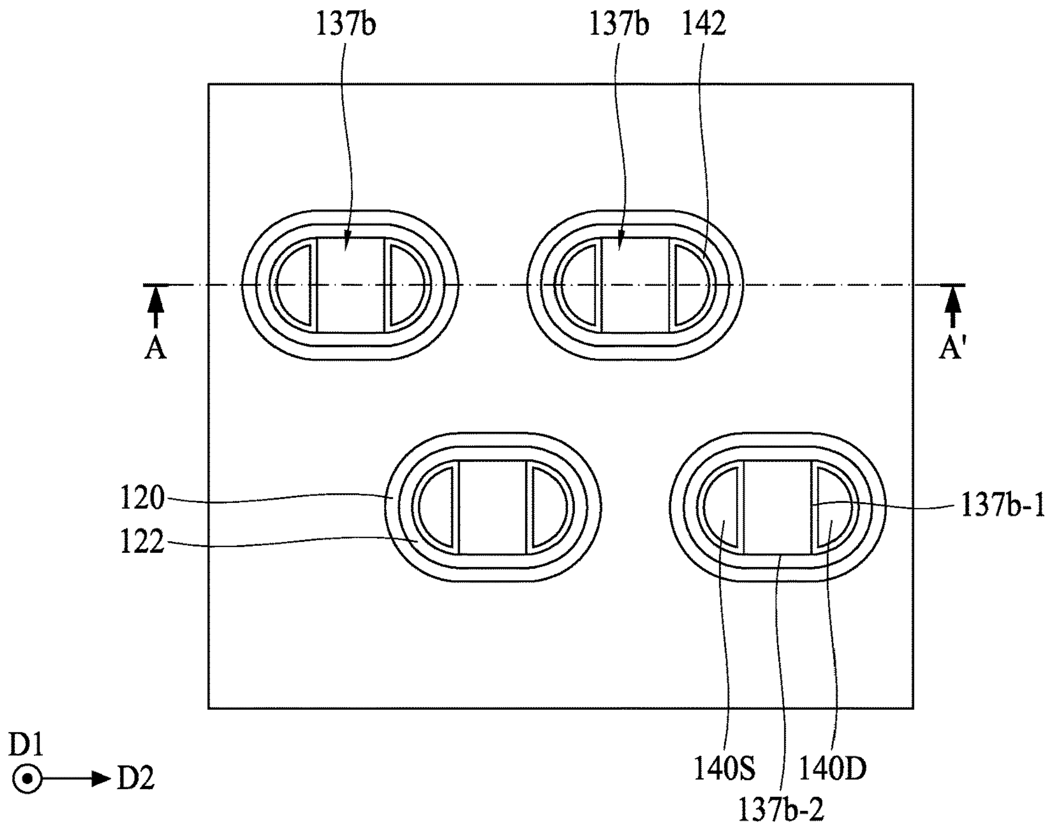


FIG. 18A

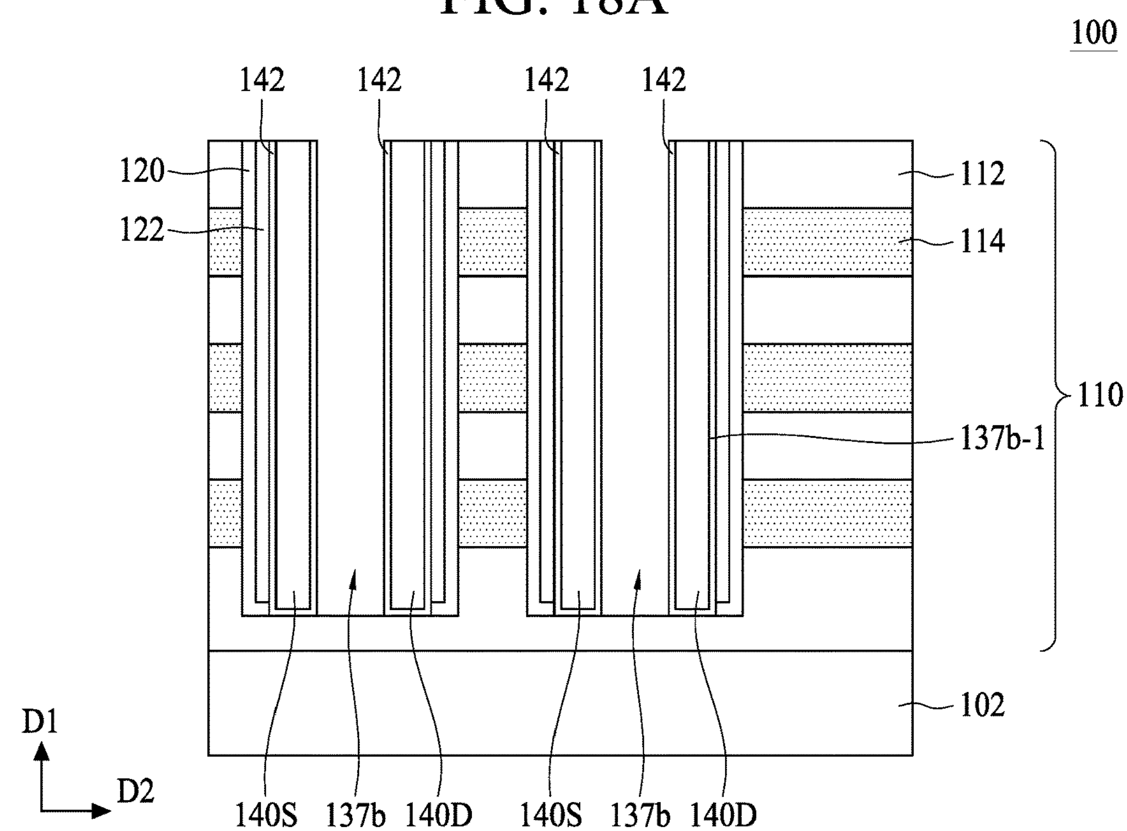


FIG. 18B

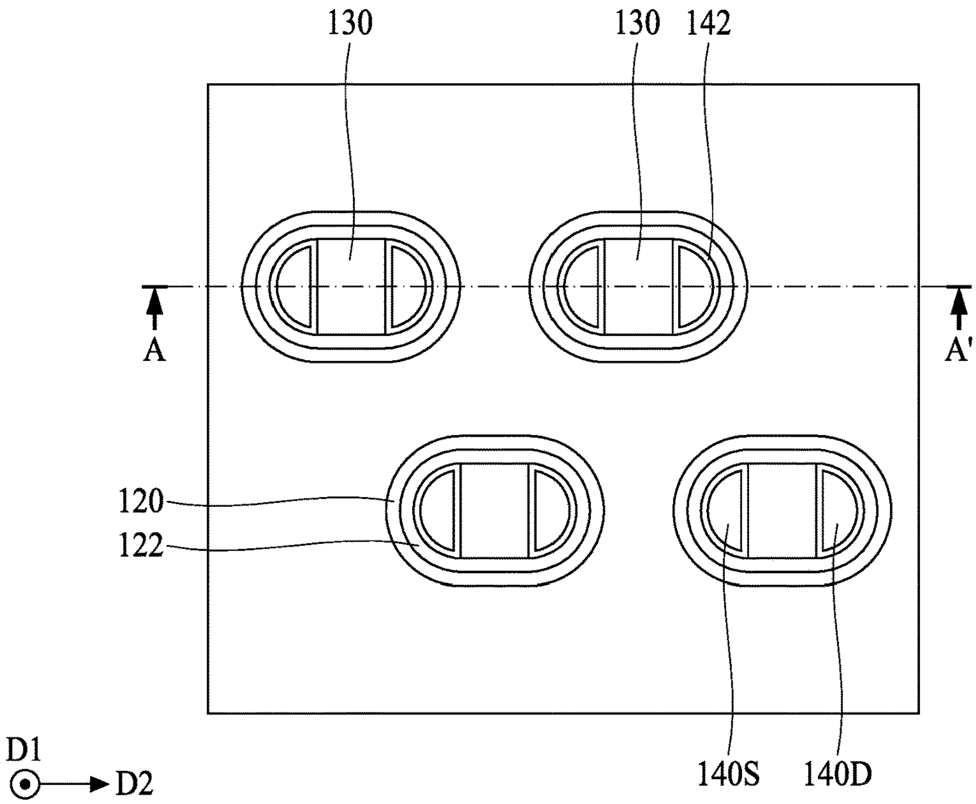


FIG. 19A

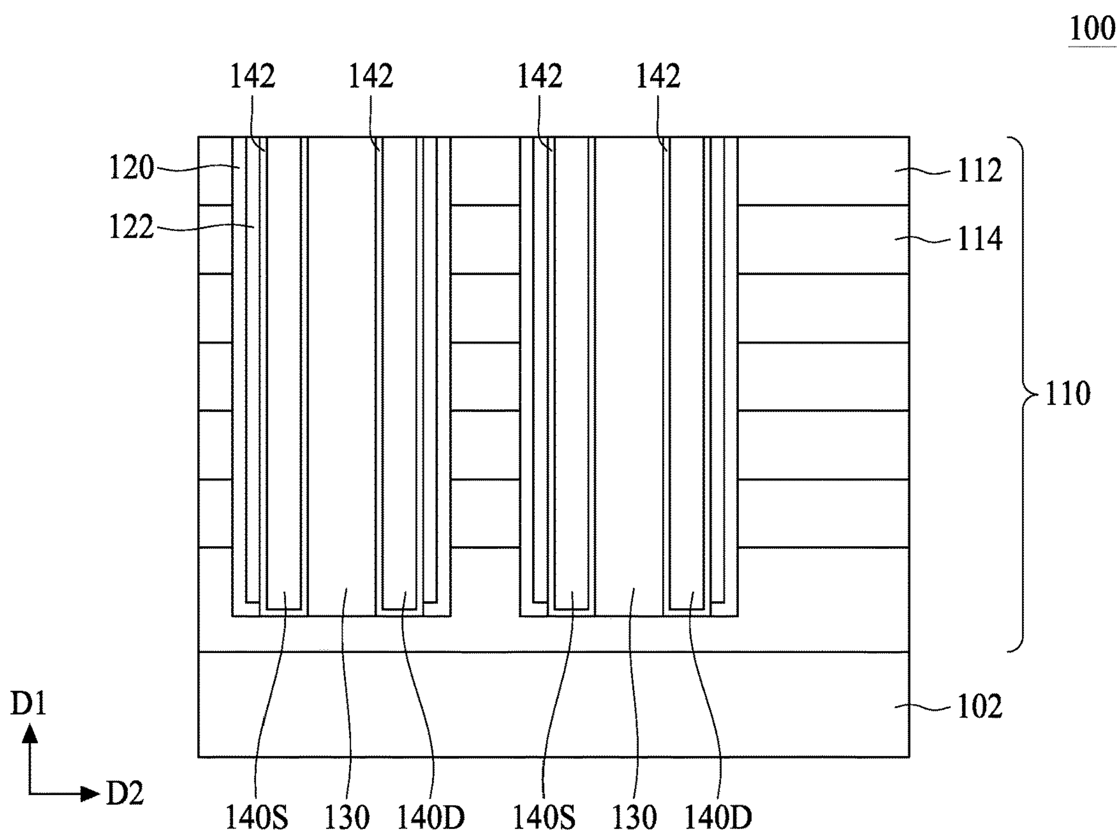


FIG. 19B

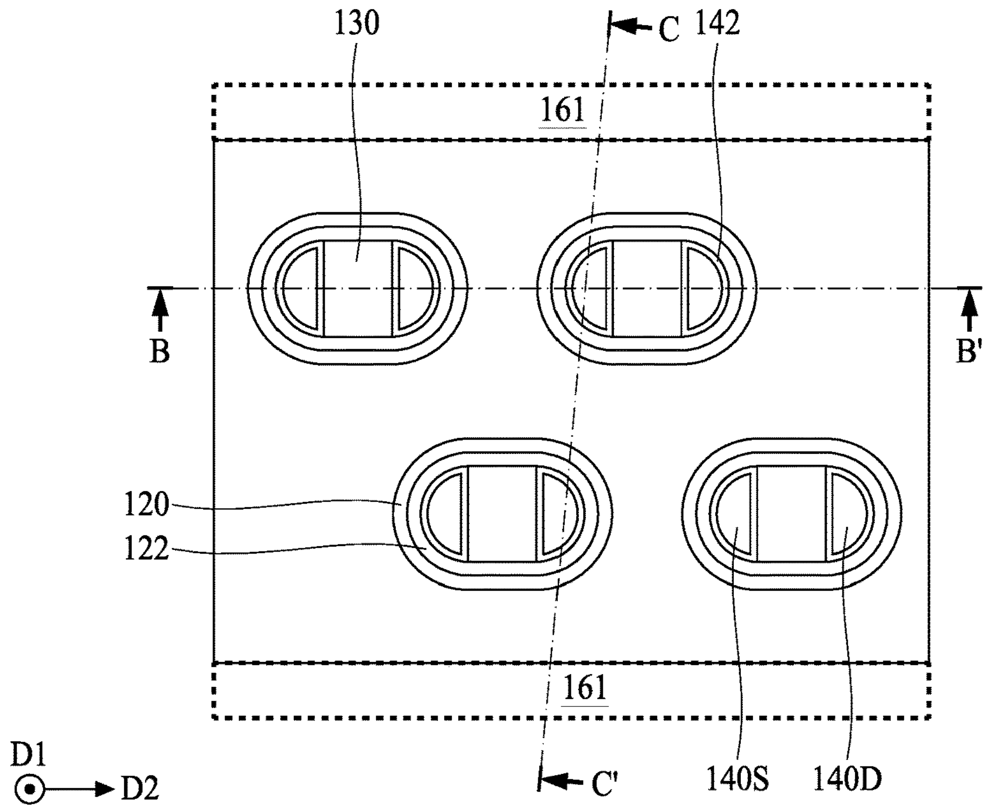
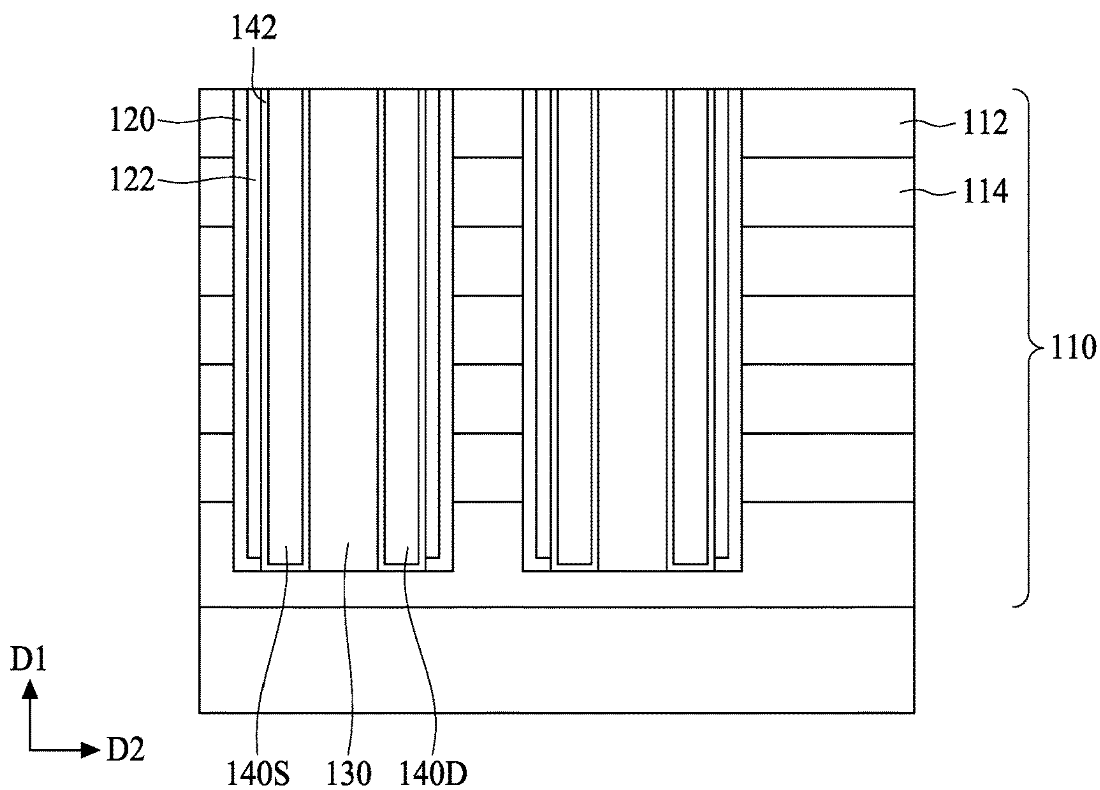


FIG. 20A



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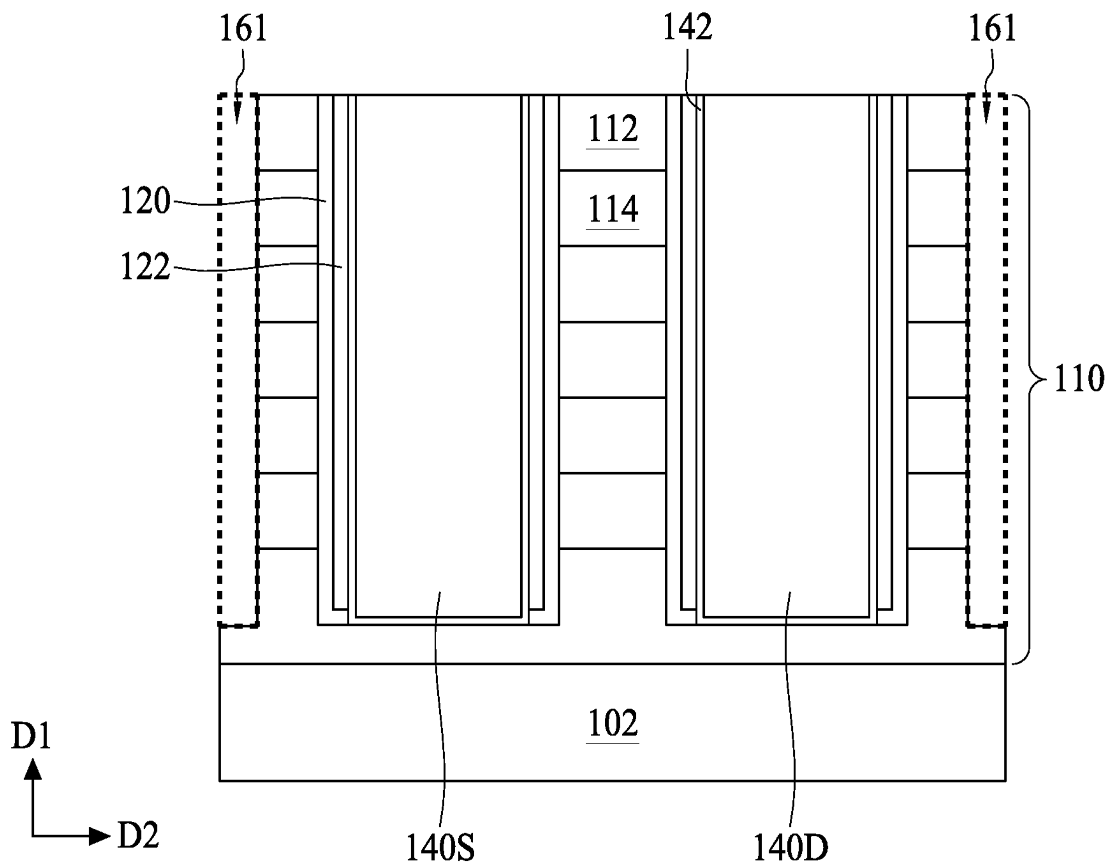


FIG. 20C

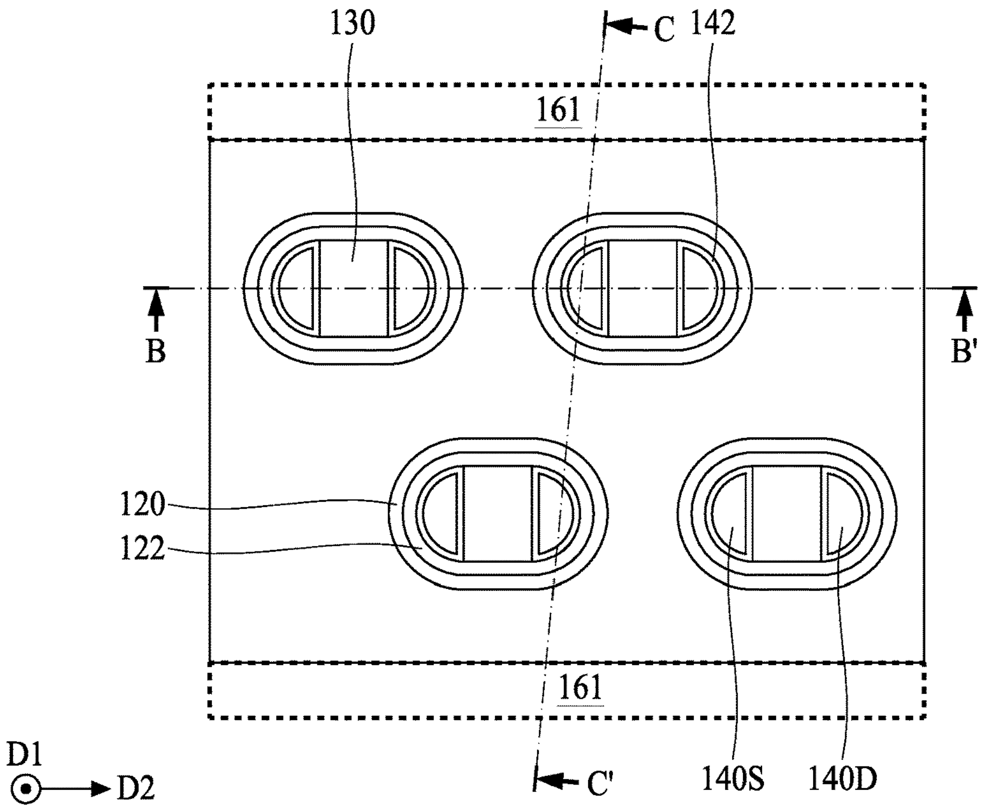


FIG. 21A

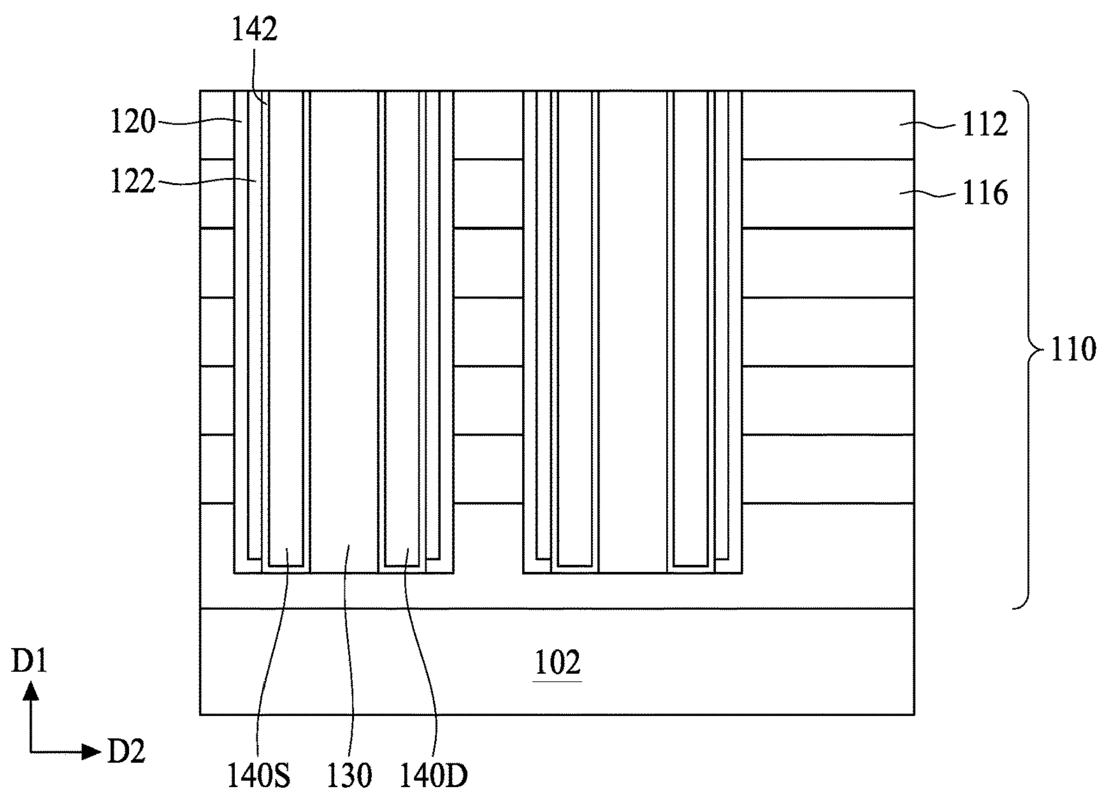


FIG. 21B

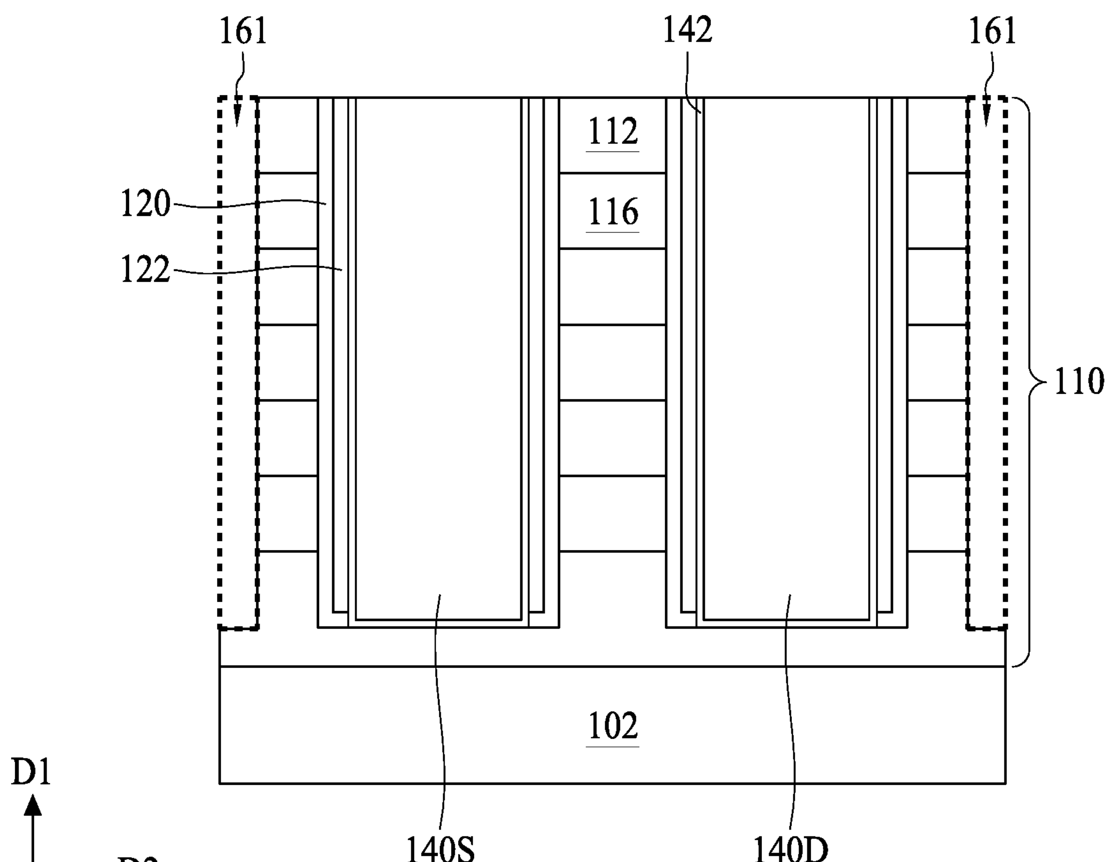


FIG. 21C

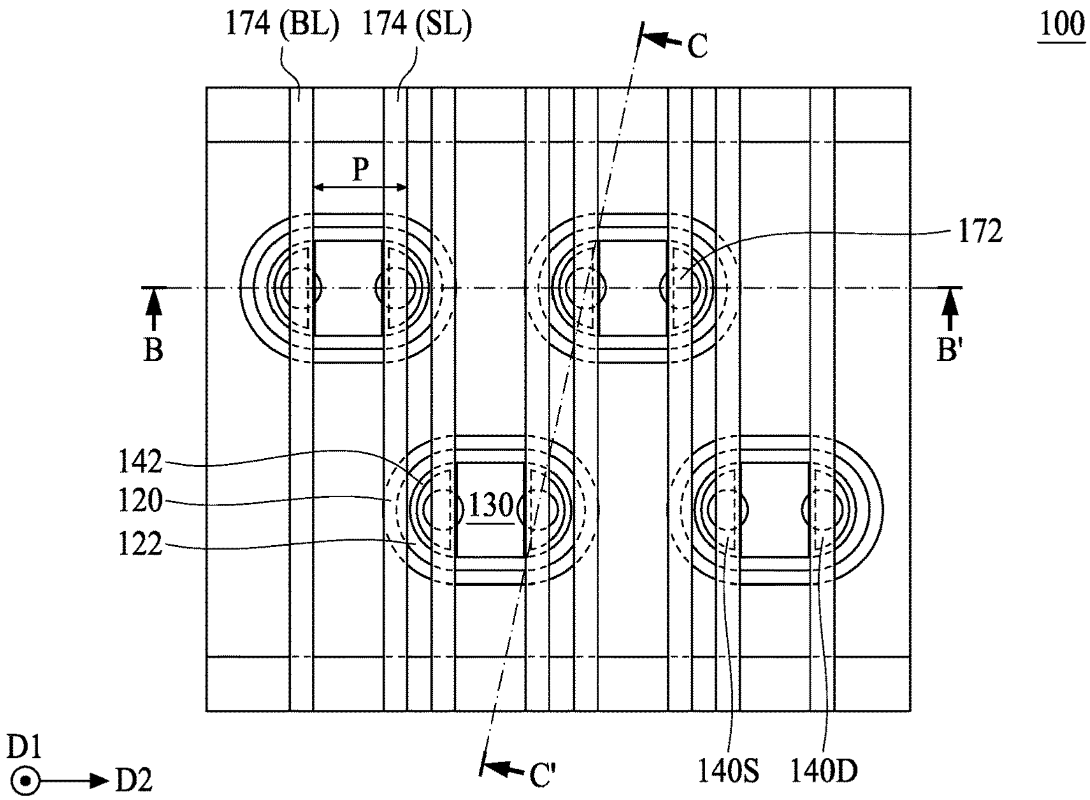


FIG. 22A

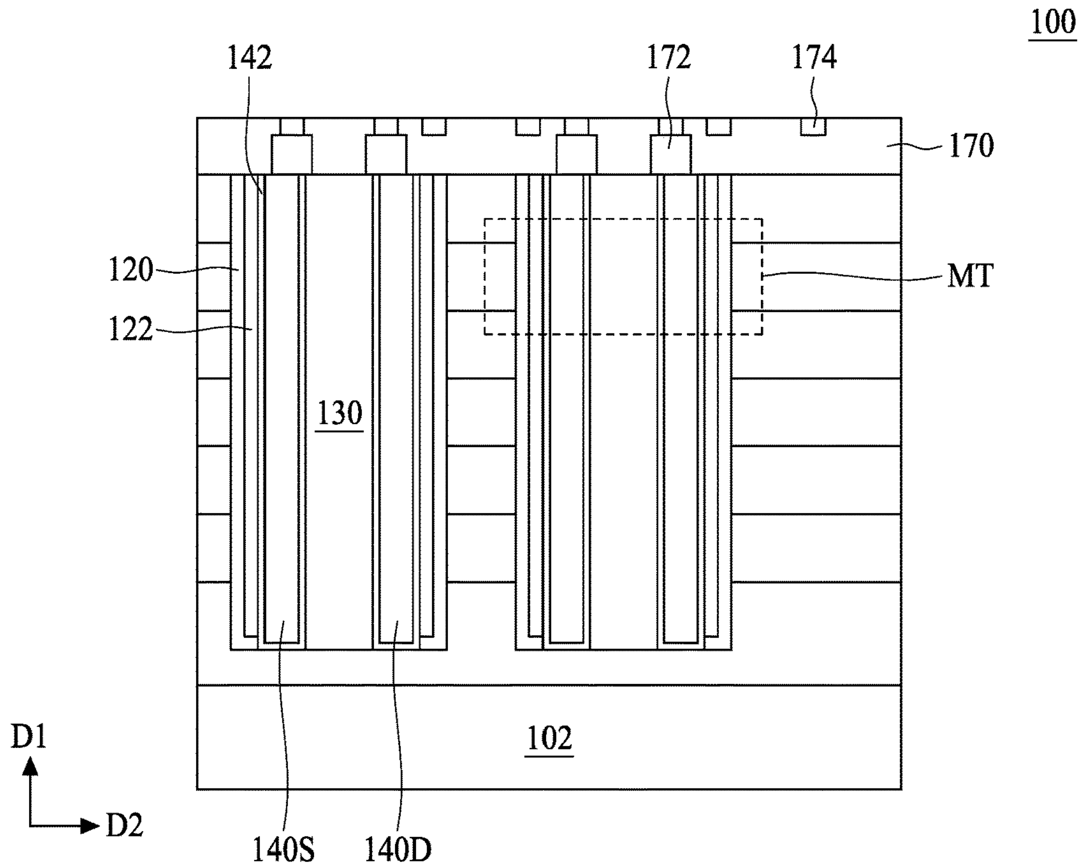


FIG. 22B

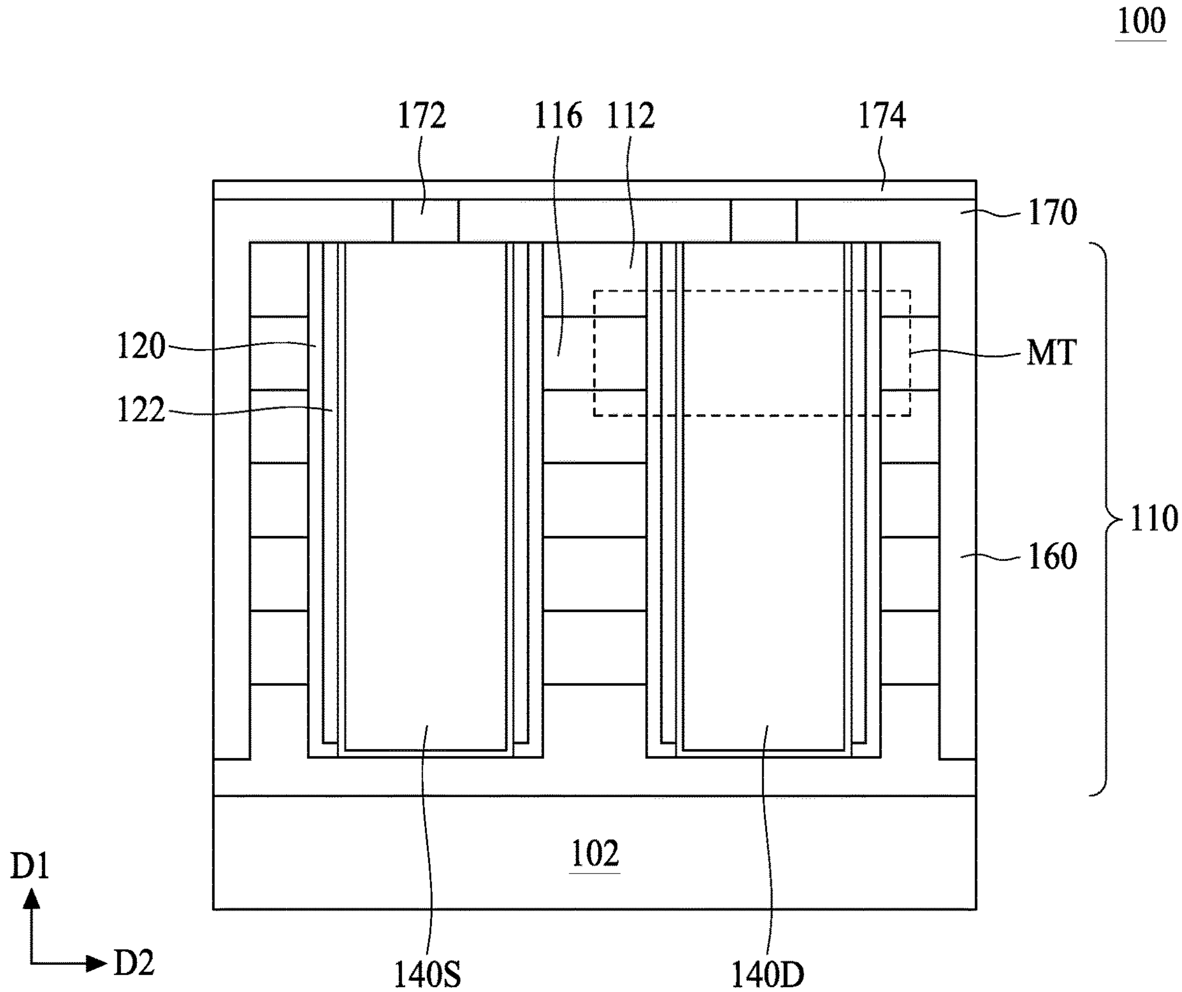
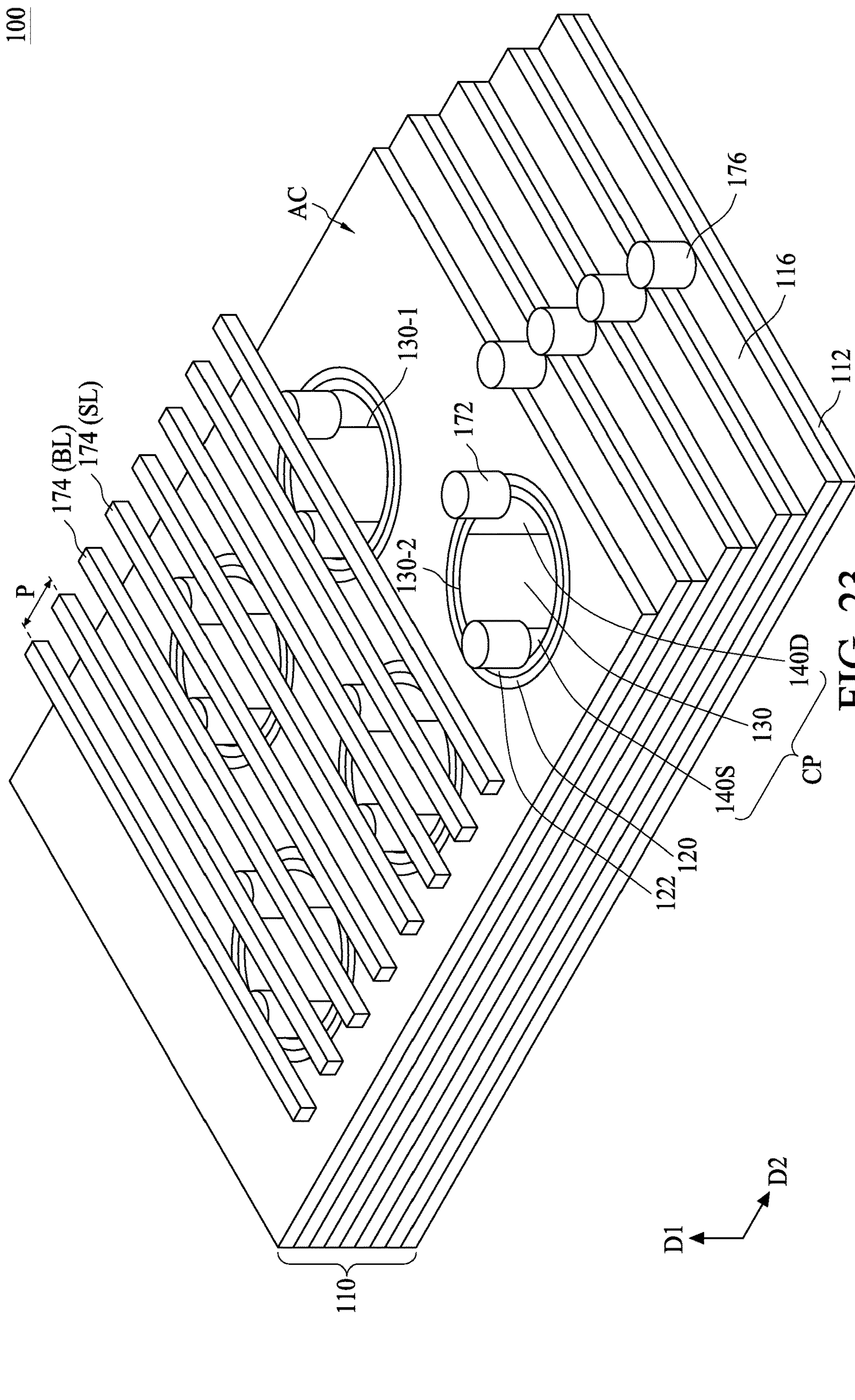


FIG. 22C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AC
130-1
174 (BL)
174 (SL)
P
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130-2
176
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140D
130
140S
CP
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122
110
D1
D2

FIG. 23

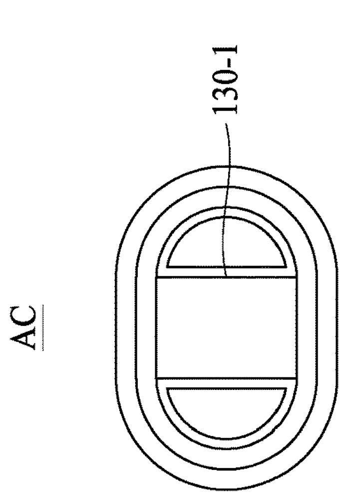

FIG. 24A

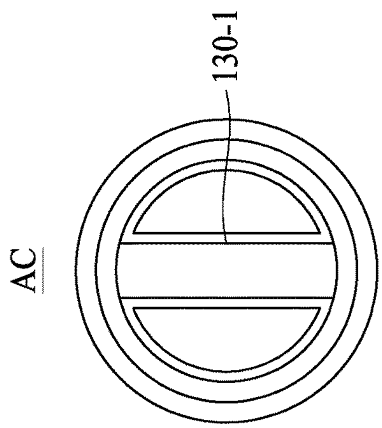

FIG. 24B

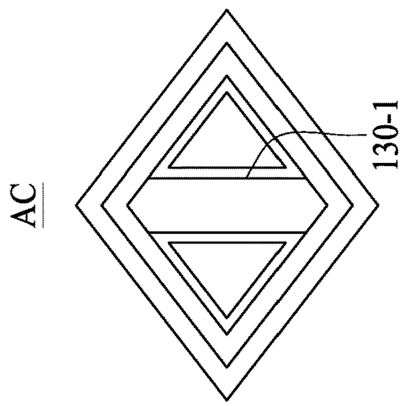

FIG. 24C

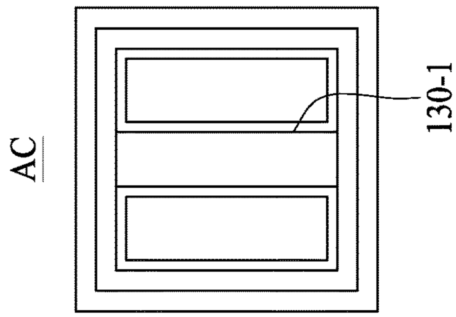

FIG. 24D

SEMICONDUCTOR MEMORY STRUCTURE AND METHOD FOR FORMING THE SAME

BACKGROUND

Recently, great progress has been achieved in development of semiconductor memory devices. Due to continuously increasing requirements for memory devices with mass capacity, integration density of memory cells in a memory device keeps increasing. Scaling the memory cell size and realizing high-density memory are eagerly needed for various applications such as internet of things (IoT) and mechanism learning. As critical dimensions of devices in integrated circuits shrink to the limits of common memory cell technologies, designers have been looking to techniques for stacking multiple planes of memory cells to achieve greater storage capacity, and to achieve lower costs per bit. A three dimensional (3D) memory array architecture has been realized by stacking several cells in series vertically up on each cell which is located in two dimensional (2D) array matrix. There is a continuous demand to develop improved semiconductor memory structures for satisfying the above requirements.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It should be noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. **1** is a flowchart of a method for forming a semiconductor memory structure according to various aspects of the present disclosure.

FIGS. **2**A, **3**A, **4**A, **5**A, **6**A, **7**A and **8**A are schematic drawings illustrating various stages in a method for forming a semiconductor memory structure according to aspects of one or more embodiments of the present disclosure.

FIGS. **2**B, **3**B, **4**B, **5**B, **6**B, **7**B and **8**B are cross-sectional views taken along line A-A' in FIGS. **2**A, **3**A, **4**A, **5**A, **6**A, **7**A and **8**A, respectively.

FIG. **9** is a flowchart of a method for forming a semiconductor memory structure according to various aspects of the present disclosure.

FIGS. **10**A, **11**A, **12**A, **13**A, and **14**A are schematic drawings illustrating various stages in a method for forming a semiconductor memory structure after performing the operation associated with FIG. **5**A according to aspects of one or more embodiments of the present disclosure.

FIGS. **10**B, **11**B, **12**B, **13**B and **14**B are cross-sectional views taken along line A-A' in FIGS. **10**A, **11**A, **12**A, **13**A and **14**A, respectively.

FIG. **15** is a flowchart of a method for forming a semiconductor memory structure according to various aspects of the present disclosure.

FIGS. **16**A, **17**A, **18**A and **19**A are schematic drawings illustrating various stages in a method for forming a semiconductor memory structure after performing the operation associated with FIG. **10**A according to aspects of one or more embodiments of the present disclosure.

FIGS. **16**B, **17**B, **18**B and **19**B are cross-sectional views taken along line A-A' in FIGS. **16**A, **17**A, **18**A and **19**B, respectively.

FIGS. **20**A, **21**A, **22**A and **23** are schematic drawings illustrating various stages in a method for forming a semiconductor memory structure according to aspects of one or more embodiments of the present disclosure.

FIGS. **20**B, **21**B and **22**B are cross-sectional views taken along line B-B' in FIGS. **20**A, **21**A and **22**A, respectively.

FIGS. **20**C, **21**C and **22**C are cross-sectional views taken along line C-C' in FIGS. **20**A, **21**A and **22**A, respectively.

FIGS. **24**A to **24**D are schematic drawings illustrating shapes of active columns according to aspects of one or more embodiments of the present disclosure.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of elements and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as "beneath," "below," "lower," "above," "upper," "on" and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The device may be otherwise oriented (rotated 100 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

As used herein, the terms such as "first," "second" and "third" describe various elements, components, regions, layers and/or sections, but these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another. The terms such as "first," "second" and "third" when used herein do not imply a sequence or order unless clearly indicated by the context.

Non-volatile memory (NVM) generally refers to any memory or storage that can retain stored data even when no power is applied. Example NVM includes flash memory, and flash memory includes two main types: NAND-type flash memory and NOR-type flash memory. The NAND-type flash memory has an advantage of a high-density structure including a number of storage transistors series-connected to form strings. However, reading and writing the content of any of the series-connected storage transistor requires activation of all series-connected storage transistors in one string, results in slow read/write access speed. Therefore, a memory structure complying with high-speed requirement in AI and high bandwidth memory, and high density requirements is needed.

The present disclosure therefore provides a semiconductor memory structure and a method for forming the same that complies with both the high read/write access speed requirement and the high density requirement. In some embodiments, the semiconductor memory structure in a NOR-type flash memory structure that provides greater read/write access speed as good as a dynamic random access memory (DRAM). Further, the semiconductor memory structure includes a 3D configuration that complies with the high-density requirement as good as NAND-type memory.

FIG. **1** is a flowchart representing a method for forming a semiconductor memory structure **10** according to aspects of the present disclosure. In some embodiments, the method for forming the semiconductor memory structure **10** includes a number of operations (**11**, **12**, **13**, **14** and **15**). The methods for forming the semiconductor memory structure **10** will be further described according to one or more embodiments. It should be noted that the operations of the methods for forming the semiconductor memory structure **10** may be rearranged or otherwise modified within the scope of the various aspects. It should further be noted that additional processes may be provided before, during, and after the method **10**, and that some other processes may be only briefly described herein.

FIGS. **2**A, **3**A, **4**A, **5**A, **6**A, **7**A and **8**A are schematic drawings illustrating various stages in a method for forming a semiconductor memory structure according to aspects of one or more embodiments of the present disclosure, and FIGS. **2**B, **3**B, **4**B, **5**B, **6**B, **7**B and **8**B are cross-sectional views taken along line A-A' in FIGS. **2**A, **3**A, **4**A, **5**A, **6**A, **7**A and **8**A, respectively. Referring to FIGS. **2**A and **2**B, in some embodiments, a substrate **102** can be received or provided. In some embodiments, the substrate **102** is a silicon substrate. Alternatively or additionally, the substrate **102** includes germanium, an alloy semiconductor (for example, SiGe), another suitable semiconductor material, or combinations thereof. Alternatively, the substrate **102** is a semiconductor-on-insulator substrate, such as a silicon-on-insulator (SOI) substrate, a silicon germanium-on-insulator (SGOI) substrate, or a germanium-on-insulator (GOI) substrate. Semiconductor-on-insulator substrates can be fabricated using separation by implantation of oxygen (SIMOX), wafer bonding, and/or other suitable methods. In some embodiments, the substrate **102** can include various doped regions (not shown) depending on design requirements of semiconductor memory structure. In some embodiments, the substrate **102** includes p-type doped regions (for example, p-type wells) doped with p-type dopants, such as boron (for example, BF2), indium, other p-type dopant, or combinations thereof. In some embodiments, the substrate **102** includes n-type doped regions (for example, n-type wells) doped with n-type dopants, such as phosphorus, arsenic, other n-type dopant, or combinations thereof. In some embodiments, the substrate **102** includes doped regions formed with a combination of p-type dopants and n-type dopants. The various doped regions can be formed directly on and/or in the substrate **102**, for example, providing a p-well structure, an n-well structure, a dual-well structure, a raised structure, or combinations thereof. Doping may be implemented using a process such as ion implantation or diffusion in various steps and techniques.

In operation **11**, a stack **110** is formed over the substrate **102**. As shown in FIGS. **2**A and **2**B, the stack **110** includes a plurality of first layers **112** and a plurality of second layers **114**. Further, the first layers **112** and the second layers **114** are alternately arranged. The number of the alternating layers included in the stack **110** can be made as high as the number of layers needed for the semiconductor memory device. Further, in some embodiments, the topmost layer and the bottom most layer can both be the first layers **112**, as shown in FIG. **2**B. In some embodiments, a thickness of the first layers **112** can be between approximately 250 Angstroms and approximately 1000 Angstroms, and a thickness of the second layers **114** can be between approximately 250 Angstroms and approximately 1000 Angstroms. It should be noted that the thickness of the first layers **112** and the thickness of the second layers **114** can be similar or different, depending on different product requirements. In some embodiments, the thickness of the first layer **112** and the thickness of the second layer **114** have a ratio, and the ratio is between approximately 0.1 and approximately 10. In some embodiments, the first layers **112** include a first insulating material, and the second layers **114** include a second insulating material different from the first insulating material. For example, in some embodiments, the first layers **112** include silicon oxide, and the second layers **114** include silicon nitride, but the disclosure is not limited thereto. In some alternative embodiments, the first layers **112** can include an insulating material while the second layers **114** can include a semiconductor material. For example, in some embodiments, the first layers **112** include silicon oxide, and the second layers **114** include silicon, but the disclosure is not limited thereto.

Referring to FIGS. **3**A and **3**B, in operation **12**, at least a trench **115** is formed in the stack **110**. A number of the trench **115** can be made as high as the number of they needed. A shape of the trench **115** can be made as rectangular, rhombus, oval, or circular, according to different product or design requirements. In some embodiments, a diameter of the trench **115** can be between approximately 100 nanometers and approximately 500 nanometers, but the disclosure is not limited thereto. As shown in FIG. **3**A, the trenches **115** can be arranged in an array configuration of rows and columns or in a staggered array configuration. As shown in FIG. **3**B, the trench **115** penetrates the stack **110**. Therefore, the alternating first and second layers **112** and **114** can be observed from sidewalls of the trench **115**. Further, the bottommost first layer **112** is exposed through a bottom of the trench **115**.

Referring to FIGS. **4**A and **4**B, in operation **13**, a charge-trapping layer **120** and a channel layer **122** are formed in the trench **115**. In some embodiments, the charge-trapping layer **120** can be referred to as a memory layer. The charge-trapping layer **120** and the channel layer **122** can be conformally formed in the trench **115** by, for example but not limited thereto, a deposition. Therefore, the charge-trapping layer **120** and the channel layer **122** cover the sidewalls and the bottom of the trench **115**. In some embodiments, a thickness of the charge-trapping layer **120** is between approximately 5 nanometers and approximately 30 nanometers, but the disclosure is not limited thereto. In some embodiments, a thickness of the channel layer **122** is between approximately 5 nanometers and approximately 30 nanometers, but the disclosure is not limited thereto. The charge-trapping layer **120** can include an insulating structure or a ferroelectric material. For example, in some embodiments, the charge-trapping layer **120** can include an insulating structure such as an oxide-nitride-oxide (ONO) structure. In some alternative embodiments, the charge-trapping layer **120** can include a ferroelectric material such as hafnium silicates (HfSIO), hafnium zirconium oxide (HfZrO, also referred to as HZO), and the like. The channel layer **122** can include semiconductor materials, oxide semiconductor (OS) materials, or 2D materials. For example, the channel layer **122** can include polysilicon, amorphous silicon, and the like. In other example, the oxide semiconductor materials can include zinc oxide, cadmium oxide, indium oxide, and the like. In still another example, the 2D materials can include graphene, but the disclosure is not limited thereto.

Referring to FIGS. **5**A and **5**B, in some embodiments, a blank etching operation can be performed to remove portions of the charge-trapping layer **120** and the channel layer **122**. In some embodiments, the portions of the charge-trapping layer **120** and the channel layer **122** covering a top surface of the stack **110** and the bottom of the trench **115** are removed. Consequently, the top surface of the stack **110** and the bottom of the trench **115** are exposed. In other words, a top surface of the topmost first layer **112** and a portion of the bottommost first layer **112** are exposed.

Referring to FIGS. **6**A and **6**B, in operation **14**, an isolation structure **130** is formed in the trench **115**. In some embodiments, the isolation structure **130** can include dielectric materials such as silicon oxide, silicon oxycarbide, silicon oxynitride, silicon carbonitride, and the like. In some embodiments, the trench **115** is filled with the dielectric material, and a planarization operation such as a CMP operation is performed to remove superfluous dielectric material. Accordingly, a top surface of the isolation structure **130** and the top surface of the stack **110** (i.e., the top surface of the topmost first layer **112**) can be aligned with each other.

In operation **15**, a source structure **140**S and a drain structure **140**D are formed at two sides of the isolation structure **130**.

Referring to FIGS. **7**A and **7**B, portions of the isolation structure **130** are removed. Accordingly, a plurality of recesses **131** are formed. In some embodiments, two recesses **131** are formed at two sides of the isolation structure **130**, as shown in FIGS. **7**A and **7**B. Further, the channel layer **122** may be exposed though sidewalls of the recesses **131**. Referring to FIGS. **8**A and **8**B, a barrier layer **142** can be formed to cover sidewalls and a bottom of each recess **131**, and then the recesses **131** are filled with a conductive material. A planarization operation such as a CMP operation is performed to remove superfluous barrier layer and conductive material. Consequently, the source structure **140**S and the drain structure **140**D are formed at the two sides of the isolation structure **130**. In some embodiments, the barrier layer **142** includes titanium, titanium nitride, tantalum nitride, ruthenium, and the like. In some embodiments, the conductive material can include doped polysilicon, doped amorphous silicon, tungsten, copper, and the like.

In some embodiments, when the second layers **114** include the semiconductor material, the second layers **114** can serve as gate layers for the semiconductor memory structure **100**, as shown in FIGS. **8**A and **8**B. Additionally, operations for forming conductive features such as conductive vias and conductive lines can be performed after the forming of the source structure **140**S and the drain structure **140**D.

According to the method for forming the semiconductor memory structure **10**, a 3D semiconductor memory structure **100** can be obtained.

FIG. **9** is a flowchart representing a method for forming a semiconductor memory structure **20** according to aspects of the present disclosure. In some embodiments, the method for forming the semiconductor memory structure **10** includes a number of operations (**21**, **22**, **23**, **24**, **25**, **26**, **27** and **28**). The methods for forming the semiconductor memory structure **20** will be further described according to one or more embodiments. It should be noted that the operations of the methods for forming the semiconductor memory structure **20** may be rearranged or otherwise modified within the scope of the various aspects. It should further be noted that additional processes may be provided before, during, and after the method **20**, and that some other processes may be only briefly described herein.

FIGS. **2**A, **3**A, **4**A, **5**A, **10**A, **11**A, **12**A, **13**A and **14**A are schematic drawings illustrating various stages in a method for forming a semiconductor memory structure according to aspects of one or more embodiments of the present disclosure, and FIGS. **2**B, **3**B, **4**B, **5**B, **10**B, **11**B, **12**B, **13**B and **14**B are cross-sectional views taken along line A-A' in FIGS. **2**A, **3**A, **4**A, **5**A, **10**A, **11**A, **12**A, **13**A and **14**A, respectively. Further, operations shown in FIGS. **10**A to **14**A and FIGS. **10**B to **14**B may be performed after performing operations associated with FIGS. **2**A to **5**A and **2**B and **5**B. It should be noted that same elements are indicated by the same numerals, and repeated descriptions of those elements can be omitted for brevity.

Referring to FIGS. **2**A and **2**B, in some embodiments, a substrate **102** can be received or provided. In some embodiments, the substrate **102** can include various doped regions (not shown) depending on design requirements of semiconductor memory structure. In operation **21**, a stack **110** is formed over the substrate **102**. As shown in FIGS. **2**A and **2**B, the stack **110** includes a plurality of first layers **112** and a plurality of second layers **114**. Further, the first layers **112** and the second layers **114** are alternately arranged. The number of the alternating layers included in the stack **110** can be made as high as the number of layers needed for the semiconductor memory device. Further, in some embodiments, the topmost layer and the bottom most layer can be the first layers **112**, as shown in FIG. **2**B. A thickness and materials used to form the first layers **112** can be similar as mentioned above, and a thickness and material used to form the second layers **114** can be similar as mentioned above, therefore those details are omitted for brevity.

Referring to FIGS. **3**A and **3**B, in operation **22**, at least a trench **115** is formed in the stack **110**. A number of the trench **115** can be made as high as the number of they needed. A shape and a diameter of the trench **115** can be can be similar as mentioned above; therefore, those details are omitted for brevity. As shown in FIG. **3**A, the trenches **115** can be arranged in an array configuration of rows and columns or in a staggered array configuration. As shown in FIG. **3**B, the trench **115** penetrates the stack **110**. Therefore, the alternating first and second layers **112** and **114** can be observed from sidewalls of the trench **115**. Further, the bottom most first layer **112** is exposed though a bottom of the trench **115**, as shown in FIG. **3**B. Referring to FIGS. **4**A and **4**B, in operation **23**, a charge-trapping layer **120** and a channel layer **122** are formed in the trench **115**. As mentioned above, the charge-trapping layer **120** can be referred to as a memory layer. Materials and methods for forming the charge-trapping layer **120** and the channel layer **122** can be similar as mentioned above, therefore those details are omitted for brevity. Referring to FIGS. **5**A and **5**B, a blank etching operation can be performed to remove portions of the charge-trapping layer **120** and the channel layer **122**. In some embodiments, the portions of the charge-trapping layer **120** and the channel layer **122** covering a top surface of the stack **110** and the bottom of the trench **115** are removed. Consequently, the top surface of the stack **110** and the bottom of the trench **115** are exposed. In other words, a top surface of the topmost first layer **112** and a portion of the bottom most first layer **112** are exposed.

Referring to FIGS. **10**A and **10**B, in operation **24**, the trench **115** is filled with a sacrificial layer **133**. In some embodiments, the sacrificial layer **133** can include SiN, Si, Polymer, Spin-on-Carbon, etc . . . . In some embodiments, the sacrificial layer **133** may cover the top surface of the stack **110**. A planarization operation such as a CMP operation can be performed to remove superfluous sacrificial layer **133**. Accordingly, a top surface of the sacrificial layer **133** and the top surface of the stack **110** (i.e., the top surface of the topmost first layer **112**) can be aligned with each other.

Referring to FIGS. **11**A and **11**B, in operation **25**, a recess **135***a* can be formed in the sacrificial layer **133** by removing a portion of the sacrificial layer **133**. In some embodiments, the recess **135***a* may include a pair of first sidewalls **135***a***-1** and a pair of second sidewalls **135***a***-2**. In some embodiments, the sacrificial layer **133** may be exposed through the pair of first sidewalls **135***a***-1** while the channel layer **122** may be exposed through the pair of second sidewalls **135***a***-2**, as shown in FIG. **11**A. In some embodiments, the first layer **112** of the stack **110**, such as the bottommost first layer **112** of the stack **110** may be exposed through a bottom of the recess **135***a*, as shown in FIG. **11**B.

Referring to FIGS. **12**A and **12**B, in operation **26**, an isolation structure **130** is formed in the recess **135***a*. In some embodiments, the recess **135***a* is filled with a dielectric material, and a planarization operation such as CMP operation is performed to remove superfluous dielectric material and to form the isolation structure **130**. Accordingly, a top surface of the isolation structure **130** and the top surface of the stack **110** (i.e., the top surface of the topmost first layer **112**) can be aligned with each other. Additionally, materials for forming the isolation structure **130** can be similar as mentioned above; therefore, those details are omitted for brevity.

Referring to FIGS. **13**A and **13**B, in operation **27**, recesses **137***a* are formed at two sides of the isolation structure **130**. In some embodiments, the remaining sacrificial layer **133** are removed, thus recesses **137***a* are formed at the two sides of the isolation structure **130**. Further, the channel layer **122** and the isolation structure **130** may be exposed though sidewalls of each recess **137***a*, while the first layer **112** of the stack **110** (i.e., the bottom most first layer **112** of the stack **110**) may be exposed through a bottom of each recess **137***a*. In some embodiments, the recesses **137***a* include a similar size.

In operation **28**, a source structure **140**S and a drain structure **140**D are respectively formed in the recesses **137***a* at the two sides of the isolation structure **130**. Referring to FIGS. **14**A and **14**B, in some embodiments, a barrier layer **142** can be formed to cover sidewalls and a bottom of each recess **137***a*, and then the recesses **137***a* are filled with a conductive material. A planarization operation such as a CMP operation is performed to remove superfluous barrier layer and conductive material. Consequently, the source structure **140**S and the drain structure **140**D are formed at the two sides of the isolation structure **130**. Materials for forming the barrier layer **142**, the source structure **140**S and the drain structure **140**D can be similar as mentioned above; therefore, those details are omitted for brevity.

In some embodiments, when the second layers **114** include the semiconductor material, the second layers **114** can serve as gate layers for the semiconductor memory structure **100**, as shown in FIGS. **14**A and **14**B. Additionally, operations for forming conductive features such as conductive vias and conductive lines can be performed after the forming of the source structure **140**S and the drain structure **140**D.

According to the method for forming the semiconductor memory structure **20**, a 3D semiconductor memory structure **100** can be obtained.

FIG. **15** is a flowchart representing a method for forming a semiconductor memory structure **30** according to aspects of the present disclosure. In some embodiments, the method for forming the semiconductor memory structure **30** includes a number of operations (**31**, **32**, **33**, **34**, **35**, **36**, **37** and **38**). The methods for forming the semiconductor memory structure **30** will be further described according to one or more embodiments. It should be noted that the operations of the methods for forming the semiconductor memory structure **30** may be rearranged or otherwise modified within the scope of the various aspects. It should further be noted that additional processes may be provided before, during, and after the method **30**, and that some other processes may be only briefly described herein.

FIGS. **2**A, **3**A, **4**A, **5**A, **10**A, **16**A, **17**A, **18**A and **19**A are schematic drawings illustrating various stages in a method for forming a semiconductor memory structure according to aspects of one or more embodiments of the present disclosure, and FIGS. **2**B, **3**B, **4**B, **5**B, **10**A, **16**B, **17**B, **18**B and **19**B are cross-sectional views taken along line A-A' in FIGS. **2**A, **3**A, **4**A, **5**A, **10**A, **16**A, **17**A, **18**A and **19**A, respectively. Further, operations shown in FIGS. **16**A to **19**A and FIGS. **16**B to **19**B may be performed after performing operations associated with FIGS. **10**A and **10**B, and same elements are indicated by the same numerals, and repeated descriptions of those elements can be omitted for brevity.

Referring to FIGS. **2**A and **2**B, in some embodiments, a substrate **102** can be received or provided. In operation **31**, a stack **110** is formed over the substrate **102**. As mentioned above, the stack **110** includes a plurality of first layers **112** and a plurality of second layers **114**. Further, the first layers **112** and the second layers **114** are alternately arranged. The number of the alternating layers included in the stack **110** can be made as high as the number of layers needed for the semiconductor memory device. Further, in some embodiments, the topmost layer and the bottom most layer can be the first layers **112**, as shown in FIG. **2**B. Referring to FIGS. **3**A and **3**B, in operation **32**, at least a trench **115** is formed in the stack **110**. Referring to FIGS. **4**A and **4**B, in operation **33**, a charge-trapping layer **120** and a channel layer **122** are formed in the trench **115**. Referring to FIGS. **5**A and **5**B, a blank etching operation can be performed to remove portions of the charge-trapping layer **120** and the channel layer **122**. In some embodiments, the portions of the charge-trapping layer **120** and the channel layer **122** covering a top surface of the stack **110** and the bottom of the trench **115** are removed. Consequently, the top surface of the stack **110** and the bottom of the trench **115** are exposed. In other words, a top surface of the topmost first layer **112** and a portion of the bottom most first layer **112** are exposed.

Referring to FIGS. **10**A and **10**B, in operation **34**, the trench **115** is filled with a sacrificial layer **133**. Materials for forming the sacrificial layer **133** can be similar as mentioned above; therefore, those details are omitted for brevity.

Referring to FIGS. **16**A and **16**B, in operation **35**, portions of the sacrificial layer **133** is removed. Accordingly, recesses **135***b* can be formed in the sacrificial layer **133**. In some embodiments, the recesses **135***b* are formed at two sides of the sacrificial layer **133**. Further, the channel layer **122** and the sacrificial layer **133** may be exposed through sidewalls of each recess **135***b*.

Referring to FIGS. **17**A and **17**B, in operation **36**, a source structure **140**S and a drain structure **140**D are formed in the recesses **135***b* at the two sides of the sacrificial layer **133**. As mentioned above, a barrier layer **142** can be formed to cover sidewalls and a bottom of each recess **135***b*, then the recesses **135***b* are filled with a conductive material. A planarization operation such as a CMP operation is performed to remove superfluous barrier layer and conductive material. Consequently, the source structure **140**S and the drain structure **140**D are formed at the two sides of each sacrificial layer **133**.

Referring to FIGS. **18**A and **18**B, in operation **37**, a recess **137***b* is formed between the source structure **140**S and the drain structure **140**D. In some embodiments, the remaining sacrificial layer **133** is removed, thus the recess **137***b* are formed between the source structure **140**S and the drain structure **140**D. In some embodiments, the recess **137***b* may include a pair of first sidewalls **137***b***-1** and a pair of second sidewalls **137***b***-2**. In some embodiments, the barrier layer **142** may be exposed through the pair of first sidewalls **137***b***-1** while the channel layer **122** may be exposed through the pair of second sidewalls **137***b***-2**, as shown in FIG. **18**A. In some embodiments, the first layer **112** of the stack **110**, such as the bottom most first layer **112** of the stack **110** may be exposed through a bottom of the recess **137***b*, as shown in FIG. **18**B.

Referring to FIGS. **19**A and **19**B, in operation **38**, an isolation structure **130** is formed in the recess **137***b*. In some embodiments, the recess **137***b* is filled with a dielectric material, and a planarization operation such as CMP operation is performed to remove superfluous dielectric material. Accordingly, a top surface of the isolation structure **130** and the top surface of the stack **110** (i.e., the top surface of the topmost first layer **112**) can be aligned with each other. Materials for forming the isolation structure **130** can be similar as mentioned above; therefore, those details are omitted for brevity.

In some embodiments, when the second layers **114** include the semiconductor material, the second layers **114** can serve as gate layers for the semiconductor memory structure **100**, as shown in FIGS. **19**A and **19**B. Additionally, operations for forming conductive features such as conductive vias and conductive lines can be performed after the forming of the source structure **140**S and the drain structure **140**D.

According to the method for forming the semiconductor memory structure **30**, a 3D semiconductor memory structure **100** can be obtained. Further, the semiconductor memory structure **100** formed by the method **30** is a 3D semiconductor memory structure. Further, the recesses **137***b*, which accommodates the isolation structure **130**, can be self-aligned formed according to the method **30**.

FIGS. **20**A, **21**A and **22**A are schematic drawings illustrating various stages in a method for forming a semiconductor memory structure according to aspects of one or more embodiments of the present disclosure. FIGS. **20**B, **21**B and **22**B are cross-sectional views taken along line B-B' in FIGS. **20**A, **21**A and **22**A, respectively, and FIGS. **20**C, **21**C and **22**C are cross-sectional views taken along line C-C' in FIGS. **20**A, **21**A and **22**A, respectively. In some embodiments, FIGS. **20**A to **22**A, **20**B to **22**B and **20**C to **22**C may be performed after performing operations associated with FIGS. **8**A and **8**B, **14**A and **14**B, and **19**A and **19**B. It should be noted that same elements are indicated by the same numerals, and repeated descriptions of those elements can be omitted for brevity.

As mentioned above, the first layers **112** and the second layers **114** may include different insulating materials. In such embodiments, the second layers **114** are replaced with a conductive material.

As shown in FIGS. **20**A, **20**B and **20**C, in some embodiments, the removal of the second layers **114** may include forming a slit **161** on two sides of an array formed by the semiconductor memory structure **100**. As shown in FIG. **20**C, the slit **161** may perpendicularly penetrates the stack

110, thus the first layers **112** and the second layers **114** are exposed through sidewalls of the slit **161**. Further, the bottom most first layer **112** may be exposed through a bottom of the slit **161**. In some embodiments, an etchant which has a higher selectivity for the second layers **114** than the first layers **112** can be introduced into the slits **161**. Accordingly, the second layers **114** are removed to form recesses between the first layers, though not shown.

Referring to FIGS. **21**A, **21**B and **21**C, the recesses between the first layers **112** are filled with a conductive material to form a plurality of gate layers **116** along a direction parallel with surfaces of the first layer **112**. In some embodiments, the conductive material may include titanium nitride and tungsten (TiN/W), titanium nitride and copper (TiN/Cu), tantalum nitride and copper (TaN/Cu), cobalt and tungsten (Co/W), ruthenium (Ru), or other suitable conductive materials.

Referring to FIGS. **22**A, **22**B and **22**C, in some embodiments, the slits **161** are filled with a dielectric structure **160**. In some embodiment, another dielectric structure **170** can be formed over the substrate **102**. Materials for forming the dielectric structures **160** and **170** can be similar as mentioned above, therefore those details are omitted. Further, in some embodiments, the dielectric structures **160** and **170** can include a same dielectric material. In some alternative embodiments, the dielectric structures **160** and **170** can include different dielectric materials. In some embodiments, conductive vias **172** and conductive lines **174** are formed in the dielectric structure **170**. The conductive vias **172** and the conductive lines **174** can include titanium and titanium nitride (Ti/TiN), tantalum nitride and tungsten (TaN/W), polysilicon, or other suitable material. Further, the conductive vias **172** are coupled to the source structure **140**S and the drain structure **140**D, respectively. Accordingly, each source structure **140**S can be coupled to a conductive line **174**, which such serves as a bit line (BL), by the conductive via **172**. Each drain structure **140**D can be coupled to a conductive line **174**, which serves as a select line (SL), by the conductive via **172**.

Additionally, the gate layers **116** may serve as word lines (WLs) and may include a staircase configuration, and conductive vias **176** may be formed over each gate layer **116**, as shown in FIG. **23**.

Referring to FIGS. **22**A, **22**B, **22**C and **23**, accordingly, in some embodiments, the semiconductor memory structure **100** is provided. The semiconductor memory structure **100** includes a plurality of gate layers **116**, and a plurality of insulating layers **112** alternately stacked over a substrate **102** along a first direction **D1**. In some embodiments, the first direction **D1** is substantially perpendicular to a surface of the substrate **102**. Further, the gate layers **116** and the insulating layers **112** are extended along a second direction **D2** perpendicular to the first direction **D1**. In some embodiments, the second direction **D2** is substantially parallel with a surface of the substrate **202**. A thickness of the insulating layers **112** can be between approximately 250 Angstroms and approximately 1000 Angstroms, and a thickness of the gate layers **116** can be between approximately 250 Angstroms and approximately 1000 Angstroms. It should be noted that the thickness of the insulating layers **112** and the thickness of the gate layers **116** can be similar or different, depending on different product requirements. In some embodiments, the thickness of the insulating layer **112** and the thickness of the gate layer **114** have a ratio, and the ratio is between approximately 0.1 and approximately 10.

The semiconductor memory structure **100** further includes at least an active column AC disposed over the substrate **102** and extending along the first direction D**1**. As shown in FIGS. **22**B, **22**C and **23**, the active column AC may perpendicularly penetrates the gate layer **116** and the insulating layer **112**. However, as shown in FIGS. **22**B and **22**C, the active column AC is separated from the substrate **102** by the bottom most insulating layer **112** of the stack **110**. As mentioned above, a number of the active column AC can be made as high as the number of they needed. In some embodiments, the active columns AC can be arranged in an array configuration of rows and columns or in a staggered array configuration. Additionally, as shown in FIGS. **24**A to **24**D, a shape of the active columns AC can be made as rectangular, rhombus, oval, or circular, according to different product or design requirements.

In some embodiments, the active column AC includes a central portion CP, a charge-trapping layer **120** surrounding the central portion CP, and a channel layer **122** between the charge-trapping layer **120** and the central portion CP. Further, the central portion CP includes an isolation structure **130**, a source structure **140**S and a drain structure **140**D. Materials for forming the charge-trapping layer **120** and the channel layer **122** can be similar as mentioned above; therefore, those details are omitted for brevity.

As shown in FIG. **23**, the isolation structure **130** of the central portion CP of the active column AC may include a pair of first sides **130-1** and a pair of second side **130-2**, and the source structure **140**S and the drain structure **140**D are disposed at the pair of first sides **130-1**. In some embodiments, the first sides **130-1** of the isolation structures **130** can be parallel with each other, but the disclosure is not limited thereto. Further, the channel layer **122** is disposed at the pair of second sides **130-2** of the isolation structure **130**. In some embodiments, the channel layer **122** in contact with the pair of second sides **130-2** of the isolation structure **130**.

The semiconductor memory structure **100** further includes a plurality of conductive vias **172** coupled to the source structure **140**S and the drain structure **140**D, respectively. Further, the semiconductor memory structure **100** includes a plurality of conductive lines **174**. As mentioned above, the source structures **140**S are electrically connected to conductive lines **174** serving as bit lines BL by the conductive vias **172**, while the drain structures **140**D are electrically connected to conductive lines **174** serving as select lines SL by the conductive vias **172**. As shown in FIGS. **22**B, **22**C and **23**, heights of the conductive vias **172** are the same, and the bit lines **174**(BL) and the select lines **174**(SL) are parallel with each other and in a same layer. Further, the bit lines **174**(BL) and the select lines **174**(SL) are alternately arranged along the second direction D**2**.

In some embodiments, a shortest distance between the source structure **140**S and the drain structure **140**D can be determined by a pitch P between the bit line **174**(BL) and its neighboring select line **174**(SL). A pitch specifies a sum of the width of a feature and the space on one side of the feature separating that feature from a neighboring feature. In some embodiments, by selecting shape for the active column AC, arranging the active columns AC and rotating the active columns AC, the pitch P between the bit line **174**(BL) and the neighboring select line **174**(SL) can be adjusted to comply different product requirements and/or design requirements. In some embodiments, the pitch P between adjacent bit line **174**(BL) and the select line **174**(SL) is between approximately 20 nanometers and 500 nanometers, but the disclosure is not limited thereto.

In some embodiment, each active column AC includes a plurality of memory devices MT. Further, each memory device MT includes a gate layer **116**, and a source structure **140**S and a drain structure **140**D surrounding by that gate layer **116**. In some embodiments, it can be referred to as a gate layer **116**, a source structure **140**S and a drain structure **140**D in substantially same plane form a memory device MT, as shown in FIGS. **22**B and **22**C. Further, channels are formed in the channel layer **122** at the pair of second sides **130-2** of the isolation structure **130**. Accordingly, the memory device MT can be referred to as a dual-channel memory transistor MT.

In some embodiments, each active column AC includes a plurality of dual-channel memory transistors MT. As shown in FIGS. **22**A, **22**B and **22**C, the source structure **140**S and the drain structure **140**D respectively penetrate the stack **110**. Therefore, it can be referred to that the source structures **140**S of the dual-channel memory transistors MT in one active column AC are in series-connected, and the drain structures **140**D of the dual-channel memory transistors MT in one active column AC are in series-connected. Further, the same gate layer **116** of each dual-channel memory transistor MT in the same plane are in parallel-connected, as shown in FIG. **23**.

Consequently, the semiconductor memory structure **100** includes the plurality of active columns AC, which include the plurality of memory devices MT stacked in 3D configuration. The 3D configuration helps to increase cell densities. Further, the electrical connections allow the semiconductor memory structure **100** work as a NOR-type flash memory.

Accordingly, the present disclosure therefore provides a semiconductor memory structure and a method for forming the same that complies with both the high read/write access speed requirement and the high density requirement. In some embodiments, the semiconductor memory structure in a NOR-type flash memory structure that provides greater read/write access speed as good as a dynamic random access memory (DRAM). Further, the semiconductor memory structure includes a 3D configuration that complies with the high-density requirement as good as NAND-type memory.

In some embodiments, a semiconductor memory structure is provided. The semiconductor memory structure includes a plurality of gate layers and a plurality of insulating layers alternately stacked over a substrate, and at least an active column disposed over the substrate. The gate layers and the insulating layers are alternately stacked along a first direction. The active column extends along the first direction and penetrates the gate layer and the insulating layer. The active column includes a central portion, a charge-trapping layer surrounding the central portion, and a channel layer between the central portion and the charge-trapping layer. The central portion of the active column includes an isolation structure, a source structure and a drain structure. In some embodiments, the source structure and the drain structure are disposed at two sides of the isolation structure.

In some embodiments, a semiconductor memory structure is provided. The semiconductor memory structure includes a plurality of gate layers and a plurality of insulating layers alternately stacked over a substrate, a plurality of active columns disposed over the substrate, a plurality of first conductive lines and a plurality of second conductive lines. The gate layers and the insulating layers are stacked along a direction substantially perpendicular to a surface of the substrate. The active columns extend along the direction, and penetrate the gate layer and the insulating layers. Each of the active columns includes a central portion, a charge-trapping layer surrounding the central portion, and a channel layer between the central portion and the charge-trapping layer. The central portion of each active column includes an isolation structure, a source structure and a drain structure.

In some embodiments, the source structure and the drain structure are disposed at two sides of the isolation structure. The first conductive lines are respectively coupled to the source structures of the active columns, and the second conductive lines are respectively coupled to the drain structures of the active columns.

In some embodiments, a method for forming a semiconductor memory structure is provided. The method include following operations. A stack including a plurality of first layers and a plurality of second layer alternately arranged is formed over a substrate. A trench is formed in the stack. A charge-trapping layer ad a channel layer covering sidewalls and a bottom of the trench is formed. An isolation structure is formed in the trench. A source structure and a drain structure are formed at two sides of the isolation structure.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A semiconductor memory structure comprising:
- a plurality of gate layers and a plurality of insulating layers alternately stacked over a substrate along a first direction; and
- at least one active column disposed over the substrate and extending along the first direction, wherein the at least one active column penetrates the gate layers and the insulating layers, and comprises:
  - a central portion comprising an isolation structure, a source structure and a drain structure, wherein the source structure and the drain structure are disposed at two sides of the isolation structure;
  - a charge-trapping layer surrounding the central portion;
  - a channel layer between the charge-trapping layer and the central portion;
  - a first barrier layer between the source structure and the channel layer, wherein a bottom surface of the source structure is in contact with the first barrier layer; and
  - a second barrier layer between the drain structure and the channel layer, wherein a bottom surface of the drain structure is in contact with the second barrier layer.

2. The semiconductor memory structure of claim **1**, wherein the plurality of gate layers and the plurality of insulating layers are extended along a second direction perpendicular to the first direction, and the second direction is substantially parallel with a surface of the substrate.

3. The semiconductor memory structure of claim **1**, wherein the charge trapping layer comprises an oxide-nitride-oxide (ONO) structure or a ferroelectric structure.

4. The semiconductor memory structure of claim **1**, wherein the isolation structure of the central portion of the at least one active column comprises a pair of first sides and a pair of second sides, and the source structure and the drain structure are disposed at the pair of first sides.

5. The semiconductor memory structure of claim **4**, wherein the channel layer is disposed at the pair of second sides of the isolation structure.

6. The semiconductor memory structure of claim **1**, wherein a first thickness of the plurality of gate layers and a second thickness of the plurality of insulating layers have a ratio, and the ratio is between approximately 0.1 and approximately 10.

7. The semiconductor memory structure of claim **1**, wherein the first barrier layer is between the source structure and the isolation structure, and the second barrier layer is between drain structure and the isolation structure.

8. The semiconductor memory structure of claim **1**, further comprising a first conductive line coupled to the source structure, and a second conductive line coupled to the drain structure.

9. The semiconductor memory structure of claim **8**, wherein the first conductive line and the second conductive line are parallel with each other.

10. The semiconductor memory structure of claim **1**, wherein the first barrier layer and the second barrier layer respectively comprise a metal or a metal nitride.

11. A semiconductor memory structure comprising:
- a plurality of gate layers and a plurality of insulating layers alternately stacked over a substrate along a direction substantially perpendicular to a surface of the substrate;
- a plurality of active columns disposed over the substrate and extending along the direction, wherein each of the plurality of active columns penetrates the gate layers and the insulating layers, and comprises:
  - a central portion comprising an isolation structure, a source structure and a drain structure, wherein the source structure and the drain structure are disposed at two sides of the isolation structure;
  - a charge-trapping layer surrounding the central portion; and
  - a channel layer disposed between the charge-trapping layer and the central portion;
- a plurality of first conductive lines respectively coupled to the source structures of the plurality of active columns; and
- a plurality second conductive lines respectively coupled to the drain structures of the plurality of active columns,

wherein each of the isolation structures of the central portions of the plurality of active columns comprises a pair of first sides and a pair of second sides, and the first sides of the isolation structures are parallel with each other.

12. The semiconductor memory structure of claim **11**, wherein the plurality of first conductive lines and the plurality of second conductive lines are parallel with each other and in a same layer.

13. The semiconductor memory structure of claim **11**, wherein the plurality of first conductive lines and the plurality of second conductive lines are alternately arranged.

14. The semiconductor memory structure of claim **11**, further comprising:
- a plurality of first conductive vias electrically connecting the source structures of the plurality of active columns to the plurality of first conductive lines; and
- a plurality of second conductive vias electrically connecting the drain structures of the plurality of active columns to the plurality of second conductive lines,
- wherein a height of the plurality of first conductive vias and a height of the plurality of second conductive vias are the same.

15. The semiconductor memory structure of claim **14**, further comprising a plurality of third conductive vias respectively coupled to the plurality of gate layers, wherein heights of the third conductive vias are different from each other.

16. The semiconductor memory structure of claim **11**, wherein the source structure and the drain structure are disposed at the pair of first sides of the isolation structure, and the channel layer is disposed at the pair of second sides of the isolation structure.

17. A method for forming a semiconductor memory structure, comprising:

forming a stack comprising a plurality of first layers and a plurality of second layers alternately arranged over a substrate;

forming at least one trench in the stack;

forming a charge-trapping layer and a channel layer covering sidewalls and a bottom of the at least one trench;

filling the at least one trench with a sacrificial layer;

removing portions of the sacrificial layer to form first recesses;

forming a source structure and a drain structure in the first recesses;

removing the sacrificial layer to form a second recess between the source structure and the drain structure; and

forming an isolation structure in the second recess.

18. The method of claim **17**, wherein the plurality of first layers comprise insulating materials and the plurality of second layers comprise semiconductor materials or conductive materials.

19. The method of claim **17**, wherein the plurality of first layers comprise a first insulating material and the plurality of second layers comprise a second insulating material different from the first insulating material.

20. The method of claim **19**, further comprising replacing the plurality of second layers with conductive layers.

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